Communication and Information Acquisition in Committees*

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Abstract

Many collective decisions rely on costly information acquisition by group members, giving rise to a free-riding problem. This paper examines how pre-vote communication affects this problem. I present a model in which voters with common interests decide whether to acquire private information before voting, and compare institutions with and without an intermediate communication stage. The model identifies two opposing effects: communication raises the incentive to acquire information by allowing a member's signal to influence others' votes (persuasion effect), but lowers it by making others' information a closer substitute for one's own (substitution effect). In equilibrium, communication increases information acquisition and improves group accuracy and welfare when information is relatively expensive, but reduces them when information is relatively cheap. I test these predictions in a pre-registered laboratory experiment with three-member committees. Communication increases acquisition when information is expensive, as predicted, but does not reduce it when information is cheap, and it improves group accuracy and welfare at both cost levels. Additional evidence helps explain the missing downside: subjects recognize the persuasion effect but insufficiently account for the substitution effect, leading them to overvalue information acquisition under communication.

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1 Introduction

Many collective decisions rely on costly information acquisition by individual group members. Jurors may need to devote attention to the evidence, hiring committee members to work through a candidate's file, and panel experts to review the studies behind a policy proposal. Information acquisition has a public-good aspect in these settings: the cost of becoming informed is borne by the individual, while the benefit of a better collective decision is shared by the group. This creates a free-riding problem, and groups may acquire less information than is socially optimal (Persico, 2004; Gerardi and Yariv, 2008).

Such decisions are often preceded by communication among group members, ranging from jury deliberation to discussion before a committee vote. By allowing the exchange of privately held information, such pre-vote communication can change how dispersed information is aggregated into the collective decision (Gerardi and Yariv, 2007; Goeree and Yariv, 2011; Iaryczower et al., 2018). When information needs to be acquired at a cost, however, communication may also change the incentive to become informed in the first place, as it can alter how much each member's information matters for the collective decision. This raises questions: Does pre-vote communication alleviate or exacerbate the free-riding problem in information acquisition? And does it ultimately improve or impair collective decision quality and welfare?

This paper studies these questions theoretically and experimentally. I consider a model of majority voting with endogenous information acquisition. An odd number of members must choose between two alternatives. They have common interests, but do not know which alternative is better. Before voting, each member decides whether to pay a cost to acquire a private signal that helps identify the better alternative. I compare two otherwise identical games. In the no-communication game, members proceed directly from information acquisition to voting. In the communication game, they exchange public cheap-talk messages after information acquisition but before voting. Both cheap talk and voting admit multiple equilibria; I follow the literature in focusing on symmetric equilibria with weakly undominated voting strategies and maximal information transmission in the communication stage.

The central theoretical insight is that communication can be a double-edged sword for information acquisition. On the one hand, communication allows a member's signal to be shared and to influence others' votes, strengthening the incentive to become informed—*the persuasion effect*. On the other hand, communication makes information acquired by others more influential and hence a stronger substitute for one's own signal, weakening the incentive to become informed—*the substitution effect*. Which effect dominates depends on how much information others acquire. When others are unlikely to be informed, communication gives

one's own signal substantial scope to influence the group, and the persuasion effect dominates; when others are likely to be informed, one's own signal can readily become redundant under communication, and the substitution effect dominates. In equilibrium, how much information others acquire depends on the information cost, so the effect of communication is cost-dependent: it raises information acquisition when information is relatively expensive, but lowers it when information is relatively cheap. That is, communication mitigates free riding when the problem is severe without communication, but exacerbates it when the problem is otherwise mild.

The same two-sided pattern carries over to collective outcomes, with lower cost cutoffs: communication improves group accuracy and welfare when information is relatively expensive and reduces both when information is relatively cheap. The latter case, in particular, reveals a tension between ex post and ex ante efficiency: communication makes the group's use of acquired information ex post efficient, but this very improvement can weaken ex ante acquisition incentives enough to leave every member worse off.

These theoretical predictions are derived under full rationality and particular equilibrium selections. I therefore examine them in a pre-registered laboratory experiment. The experiment implements the voting game with costly information acquisition described above, using a 2×2 between-subject design that varies the availability of communication (NC and C) and the cost of acquiring a signal (low and high). Committees consist of three members, and signals are highly informative but imperfect. The two cost levels are chosen so that the model predicts opposite signs: communication increases information acquisition, decision quality, and welfare at the high cost, but decreases all three at the low cost.

The experiment reveals the beneficial side of communication but not its theoretically predicted downside. Communication increases information acquisition at the high cost, as predicted, whereas the predicted decrease at the low cost does not emerge. Pooling across cost levels, acquisition is significantly higher with communication, and the communication effect does not differ significantly across costs. Communication also improves collective decision quality and welfare at both cost levels, consistent with the aggregation advantage of communication and the absence of the predicted reduction in acquisition.

Additional measures suggest that the missing downside reflects subjects overvaluing acquisition under communication. Subjects' perceived value of acquisition—how much they think their own signal improves the group's chance of deciding correctly—is higher with communication than without at both cost levels, and exceeds its empirical counterpart only under communication. The contrast is sharpest at the low cost: acquisition is in fact less valuable with communication, yet subjects perceive it as more valuable. This misperception

appears to come not from how much subjects expect others to acquire, but from how they value their own acquisition conditional on others' choices. Where communication raises the empirical value of acquisition, more subjects say they would acquire with communication than without; where it lowers that value, the share is unchanged. Subjects thus appear to recognize the persuasion effect but not the substitution effect. The asymmetry admits two behavioral interpretations—under-inference from others' messages and overvaluation of the contribution of an additional signal.

The paper informs the design and evaluation of deliberative institutions. Allowing discussion prior to voting changes not only how a group aggregates the information its members possess, but also how much information members choose to acquire in the first place. Theoretically, this provides a caveat: even costless, nonbinding communication can be strictly harmful, since better aggregation of information can intensify free riding in providing it. Behaviorally, however, this downside appears weaker than the fully strategic benchmark predicts, as members may not fully internalize the free-riding incentive that communication amplifies. Together, the theory and the experiment suggest that whether deliberation is allowed is itself a lever for encouraging information provision in committees, and that evaluations of deliberation should not treat the amount of information as fixed.

Related literature. This paper contributes mainly to two strands of literature. First, it contributes to the literature on endogenous information acquisition in collective decisions by studying the role of pre-vote communication. A large theoretical literature studies how information acquisition depends on features of collective decision-making institutions, including group size (Persico, 2004; Martinelli, 2006; Gerardi and Yariv, 2008; Koriyama and Szentes, 2009), voting and more general decision rules (Li, 2001; Persico, 2004; Gerardi and Yariv, 2008; Gershkov and Szentes, 2009), and abstention rules (Oliveros, 2013; Rohde, 2024), as well as on motives such as reputation concerns (Swank and Visser, 2023). Earlier work, however, generally does not treat the availability of pre-vote communication itself as the institutional variation of interest.[1] Two contemporaneous working papers, developed independently of this paper, also study pre-vote communication and information acquisition. Domènech-Gironell et al. (2024) consider a continuous choice of signal precision and empha-

[1]Most of it assumes no pre-vote communication. A few papers accommodate information sharing or admit a communication interpretation without varying its availability: Li (2001) assumes acquired evidence is publicly observed; Koriyama and Szentes (2009) assume ex-post efficient aggregation, which they note can be implemented through communication; Gerardi and Yariv (2008) note that their optimal device can be interpreted as an optimal communication protocol; and Gershkov and Szentes (2009) study sequential mechanisms for acquiring and eliciting information. Swank and Visser (2023) model deliberation explicitly but keep it available throughout, focusing on reputation concerns.

size the emergence of asymmetric specialization,[2] while Hoshino and Gomberg (2025) study opinion polls in large elections using a static rational-inattention framework. In contrast, I build on the binary acquisition framework of Persico (2004) and identify two opposing effects of communication, yielding cost-dependent equilibrium predictions.

On the experimental side, existing studies examine endogenous costly information acquisition under variations in group size, information costs and signal precision, voting rules, and abstention (Großer and Seebauer, 2016; Bhattacharya et al., 2017; Elbittar et al., 2020; Meyer and Rentschler, 2023). These experiments do not allow pre-vote communication. This paper therefore also provides the first experimental evidence on how pre-vote communication affects information acquisition.

Second, this paper contributes to the literature on how deliberation affects information aggregation in collective decision-making. This literature studies how pre-vote communication shapes information revelation, strategic voting, and collective outcomes (Coughlan, 2000; Austen-Smith and Feddersen, 2006; Gerardi and Yariv, 2007; Guarnaschelli et al., 2000; Goeree and Yariv, 2011; Iaryczower et al., 2018).[3] Several studies show that deliberation can improve collective outcomes and attenuate differences across voting rules (Coughlan, 2000; Gerardi and Yariv, 2007; Goeree and Yariv, 2011), while Iaryczower et al. (2018) show, using data from U.S. appellate courts, that deliberation can also harm collective decisions. A common feature of this literature, however, is that private information is taken as exogenously given. By endogenizing information acquisition, I add a new dimension to this debate: deliberation affects not only how existing information is aggregated, but also how much information is acquired in the first place.

More broadly, this paper also relates to work on the interaction between communication and endogenous information acquisition in other settings. In sender–receiver games, endogenous information acquisition can alter the predictions of standard cheap-talk models (Pei, 2015; Argenziano et al., 2016). In financial markets, information sharing through social networks can crowd out private information acquisition by inducing free riding (Han and Yang, 2013; Halim et al., 2019). I study this interaction in collective decision-making, where communication can reshape information-acquisition incentives by changing the role of private information in collective decisions.

The remainder of the paper is organized as follows. Section 2 presents the theory.

[2]In their model, whenever a symmetric equilibrium with deliberation exists, individual precision is unchanged relative to its level in the symmetric equilibrium without deliberation. The contrast with the present paper stems from the acquisition margin: with continuous precision, all voters are equally informed in a symmetric equilibrium, whereas with binary acquisition, a symmetric mixed-strategy equilibrium leaves some voters better informed than others, which is what gives communication a role.

[3]See Martinelli and Palfrey (2020) for a broader survey of the experimental literature on communication and information in collective decision games.

Section 3 describes the experimental design. Section 4 reports the main experimental results. Section 5 examines the behavioral mechanisms underlying the observed acquisition patterns. Section 6 concludes.

2 Theory

2.1 Setup

A group of $N = 2n + 1$ $(n = 1, 2 \dots)$ individuals must choose an alternative $d \in \{A, B\}$ collectively. The group decision is made via majority voting, with abstention not allowed. The true state of the world $z \in \{A, B\}$ is realized at the beginning of the game, each state occurring with equal probability. Voters are uncertain about the realization of the state but share the correct prior of $\frac{1}{2}$. All voters share the same preference over the group decision: if $d = z$, each voter receives a payoff normalized to 1; otherwise each receives 0.

Each voter can pay a cost $c > 0$ to acquire a private signal $s_i \in \{a, b\}$. Signals are informative and match the state with probability q, i.e., $\Pr(s_i = a \mid z = A) = \Pr(s_i = b \mid z = B) = q > \frac{1}{2}$. Voter i's post-acquisition type is $t_i \in T := \{\emptyset, a, b\}$, where $t_i = \emptyset$ denotes non-acquisition and $t_i \in \{a, b\}$ denotes the realized signal. Both the acquisition decision and the realized signal are private information.

I consider two games. In the *no-communication game* (NC), after the information acquisition stage, each voter i casts a vote $v_i \in \{A, B\}$. Voter i's strategy consists of an information acquisition probability $\alpha_i \in [0, 1]$ and a voting strategy, i.e., a mapping from $t_i \in T$ to a probability distribution over the two votes. In the *communication game* (C), after the information acquisition stage, each voter publicly sends a cheap-talk message $m_i \in M$, after which each voter casts a vote. The message space contains at least three distinct messages, $|M| \geq 3$, so that it is rich enough to distinguish the three post-acquisition types. Voter i's strategy in game C consists of an information acquisition probability $\alpha_i \in [0, 1]$, a communication strategy that maps $t_i \in T$ into a probability distribution over M, and a voting strategy that maps his or her type and the observed message profile into a probability distribution over the two votes. For both games, let $\mu_i = \Pr(z = A \mid \mathcal{I}_i)$ denote voter i's posterior at the voting stage, where $\mathcal{I}_i = t_i$ in NC and $\mathcal{I}_i$ consists of t_i and the publicly observed message profile in C.

I focus on symmetric and neutral Perfect Bayesian equilibria (PBE). Symmetry requires that all voters use identical strategies, while neutrality requires that strategies be invariant to relabeling the two alternatives together with the corresponding signals. Both restrictions are natural here, as voters and alternatives are ex ante identical.

I impose two further restrictions to address the multiplicity of equilibria typical of voting and cheap-talk games. First, as is standard in voting games, I restrict attention to weakly undominated voting strategies. This helps rule out equilibria sustained only by the fact that a voter is never pivotal. Second, as is common in models with cheap talk (Ottaviani and Sørensen, 2001; Fehrler and Hughes, 2018; Chen et al., 2008), I focus on the most informative equilibria in game C, namely those in which public messages reveal as much about the realized post-acquisition type profile as is attainable in equilibrium. This provides a natural benchmark for assessing the potential effect of communication; at the other extreme, game C also admits babbling equilibria in which communication is uninformative and equilibrium outcomes replicate those of game NC.

Throughout, "equilibrium" refers to a symmetric and neutral PBE with weakly undominated voting strategies and, in game C, maximal information transmission. I call voting *sincere* if a voter chooses the alternative favored by his or her posterior at the voting stage: voter i votes for A whenever $\mu_i > \frac{1}{2}$ and for B whenever $\mu_i < \frac{1}{2}$; either vote is sincere when $\mu_i = \frac{1}{2}$. The lemma below characterizes post-acquisition behavior in each game.

Lemma 1 (Post-acquisition behavior)**.** *In equilibrium:*

1. *In game NC, voting is sincere. An uninformed voter randomizes uniformly between the two alternatives.*

2. *In game C, communication fully reveals the realized post-acquisition type profile and voting is sincere.*

Given sincere voting in both games and full revelation in game C, symmetric equilibria are characterized by the common acquisition probabilities $\alpha^{*NC} \in [0,1]$ and $\alpha^{*C} \in [0,1]$. In what follows, I examine how equilibrium acquisition, the quality of group decisions, and welfare differ between the two games. Closed-form expressions used in the analysis are derived in Appendix A, and proofs of all theoretical results are provided in Appendix B.

2.2 Effects of communication on incentives

Consider voter i's best response, given that all other voters acquire information with the same probability $\alpha \in [0,1]$. Let

$$\Delta^X(\alpha) := \Pr^X(d = z \mid i \text{ acquires}; \alpha) - \Pr^X(d = z \mid i \text{ does not acquire}; \alpha), \qquad X \in \{NC, C\},$$

denote her value of information acquisition.[4] Voter i's net incentive to acquire information is $\Delta^X(\alpha) - c$. Throughout this subsection, I fix α and suppress it from probability notation. In game NC, let V_A^{-i} and V_B^{-i} denote the numbers of votes for A and B, respectively, among the other $2n$ voters.[5] In both games, let S_a^{-i} and S_b^{-i} denote the numbers of acquired signals a and b, respectively, among the other $2n$ voters. Given α, the pair (S_a^{-i}, S_b^{-i}) has the same distribution in both games.

In game NC, voters vote according to their private signals if they acquire information and randomize uniformly between the two alternatives otherwise. Hence, voter i's type can affect the collective decision only through her own vote. Her information acquisition affects the probability of a correct collective decision only on the *vote-pivotal* event $V_A^{-i} = V_B^{-i}$, when the other voters' votes are tied and her vote breaks the tie. On this event, the collective decision is determined by her vote, which is correct with probability q if she acquires a signal and with probability $\frac{1}{2}$ if she remains uninformed. Therefore, the value of information acquisition in game NC is

$$\Delta^{NC}(\alpha) = (q - \tfrac{1}{2}) \Pr(V_A^{-i} = V_B^{-i}). \tag{1}$$

In game C, because of full revelation, voters share a common posterior μ after communication. If $\mu > 1/2$, voters unanimously vote for A; if $\mu < 1/2$, voters unanimously vote for B; and if $\mu = 1/2$, either collective decision may arise in equilibrium. Thus, unlike in game NC, voter i's type can affect the collective decision through its effect on the common posterior. Her information acquisition affects the probability of a correct collective decision only on the *information-pivotal* event $S_a^{-i} = S_b^{-i}$, when the other voters' acquired signals are tied so that her signal swings the group's posterior.[6] On the information-pivotal event, if voter i does not acquire information, the common posterior is $1/2$, where any collective decision gives expected $\Pr(d = z) = 1/2$. If she acquires information, her signal tilts the common posterior to strictly prefer an alternative and the group chooses the alternative indicated by her signal, yielding $\Pr(d = z) = q$. Hence, the value of information acquisition in game C is

$$\Delta^{C}(\alpha) = \left(q - \tfrac{1}{2}\right) \Pr\left(S_a^{-i} = S_b^{-i}\right). \tag{2}$$

[4] I refer to voter i using feminine pronouns in both games.

[5] In game C, full revelation implies a common posterior at the voting stage. When this posterior is strict, sincere voting is unanimous. When it equals $1/2$, vote counts are irrelevant: either collective decision is correct with probability $1/2$. Vote counts in C therefore play no role in the pivotal-event analysis beyond the signal counts, so I do not introduce separate notation for them.

[6] When $|S_a^{-i} - S_b^{-i}| = 1$, acquiring a signal can change a strict common posterior into a tie and hence may affect the collective decision, but it does not affect the probability of a correct collective decision. In particular, $\Pr(d = z \mid S_a^{-i} - S_b^{-i} = 1,\ i \text{ acquires}) = \Pr(d = z \mid S_a^{-i} - S_b^{-i} = 1,\ i \text{ does not acquire}) = q$, and the case $S_a^{-i} - S_b^{-i} = -1$ is analogous.

The difference $\Delta^C(\alpha)-\Delta^{NC}(\alpha)$ is generally non-zero because the vote-pivotal event and the information-pivotal event differ. In particular, $S_a^{-i}=S_b^{-i}$ is not equivalent to $V_A^{-i}=V_B^{-i}$ in game NC, because vote counts also include randomized votes cast by uninformed voters. To decompose this difference, let U_A^{-i} and U_B^{-i} denote the numbers of uninformed votes for A and B, respectively, among the other $2n$ voters in game NC. The vote counts in NC are thus $V_A^{-i}=S_a^{-i}+U_A^{-i}$ and $V_B^{-i}=S_b^{-i}+U_B^{-i}$. The effect of communication on the value of information acquisition can then be written as

$$\Delta^C(\alpha)-\Delta^{NC}(\alpha)=\underbrace{(q-\tfrac{1}{2})\Pr\big((S_a^{-i}=S_b^{-i})\cap(U_A^{-i}\neq U_B^{-i})\big)}_{\text{persuasion effect}}$$
$$\underbrace{-(q-\tfrac{1}{2})\Pr\big((S_a^{-i}\neq S_b^{-i})\cap(S_a^{-i}+U_A^{-i}=S_b^{-i}+U_B^{-i})\big)}_{\text{substitution effect}},$$

which consists of two terms with opposite signs, as interpreted below.

Persuasion effect. The first term captures a positive effect of communication on the value of information acquisition. Consider the information-pivotal event $S_a^{-i}=S_b^{-i}$. In game C, voter i's signal determines the group decision, so her information acquisition is valuable. In game NC, the votes of informed voters balance on this event, but voter i need not be vote-pivotal: if $U_A^{-i}\neq U_B^{-i}$, uninformed random votes tilt the aggregate vote, and voter i's vote cannot affect the collective decision.[7] Thus, *when others' acquired signals are balanced*, communication can increase the value of voter i's signal: with communication, her signal directly determines the group decision, whereas without communication noise from uninformed votes may dissipate its impact. I refer to this term as the *persuasion effect*: communication enables a voter to transmit her information and sway others' votes, amplifying the impact of her signal on the collective decision and thereby strengthening the incentive to acquire it.

For example, with $N=5$, suppose exactly two of the other four voters acquire information, one receiving signal a and the other signal b. In game C, these signals cancel out, so voter i's signal determines which alternative the common posterior favors. In game NC, however, if the two uninformed voters cast the same vote, the aggregate vote among the other four voters is not tied, so voter i's vote cannot affect the collective decision.

Substitution effect. The second term captures a negative effect of communication on the value of information acquisition. Consider a non-information-pivotal event $S_a^{-i}\neq S_b^{-i}$. In game C, voter i's information acquisition has no value: under full revelation, the group's

[7] By parity, $U_A^{-i}\neq U_B^{-i}$ implies $|U_A^{-i}-U_B^{-i}|\geq 2$ on this event.

posterior strictly favors one alternative based on others' information, and voter i's signal cannot reverse the existing informational lead. In game NC, by contrast, although informed votes are imbalanced, uninformed random votes may offset this imbalance. If $S_a^{-i} + U_A^{-i} = S_b^{-i} + U_B^{-i}$, voter i is vote-pivotal and her information acquisition has positive value. Thus, *when others' acquired signals are imbalanced*, communication can reduce the value of voter i's signal: with communication, others' information is aggregated more efficiently, which renders voter i's own signal redundant and reduces the value of acquiring it. I refer to this term as the *substitution effect*: communication makes different voters' information closer substitutes in improving group accuracy and therefore crowds out incentives to acquire it.

For example, again with $N = 5$, suppose exactly two of the other four voters acquire information and both receive signal a. In game C, if voter i acquires information, the resulting signal profile is either three a signals or two a signals and one b signal, so the common posterior favors A and the group chooses A in either case. In game NC, however, the two informed voters vote for A; if the two uninformed voters both vote for B, the other four votes are tied, making voter i's vote pivotal.

The two effects have opposite signs, and which one dominates depends on the other voters' acquisition probability α. The next lemma shows that the net effect of communication changes sign exactly once: communication raises the value of information acquisition when α is low and lowers it when α is high. Figure 1 illustrates this single-crossing pattern.

Lemma 2 (Single-crossing)**.** *For all n and q, there exists a unique cutoff $\hat{\alpha} \in (0,1)$ such that $\Delta^C(\alpha) > \Delta^{NC}(\alpha)$ for $\alpha \in [0, \hat{\alpha})$ and $\Delta^C(\alpha) < \Delta^{NC}(\alpha)$ for $\alpha \in (\hat{\alpha}, 1)$. Moreover, $\Delta^C(\hat{\alpha}) = \Delta^{NC}(\hat{\alpha})$ and $\Delta^C(1) = \Delta^{NC}(1)$.*

The intuition is as follows. When others are unlikely to acquire information ($\alpha \in [0, \hat{\alpha})$), uninformed-vote noise is substantial, so there is substantial scope for communication to prevent voter i's acquired information from being dissipated by random uninformed votes. The persuasion effect therefore dominates, and the value of information acquisition is higher in game C. At the extreme $\alpha = 0$, only the persuasion effect is operative: no other voter acquires information, so others' acquired signals can never be imbalanced and the substitution effect vanishes. By contrast, when others are relatively likely to acquire information ($\alpha \in (\hat{\alpha}, 1)$), there is less uninformed-vote noise, leaving less scope for the persuasion effect and making voter i's own signal more likely to be redundant under communication; the substitution effect dominates, and the value of information acquisition is lower in game C.[8]

[8]The magnitude of the negative gap need not increase monotonically with α. The substitution effect requires both an imbalance in others' acquired signals, which can make voter i's signal redundant under

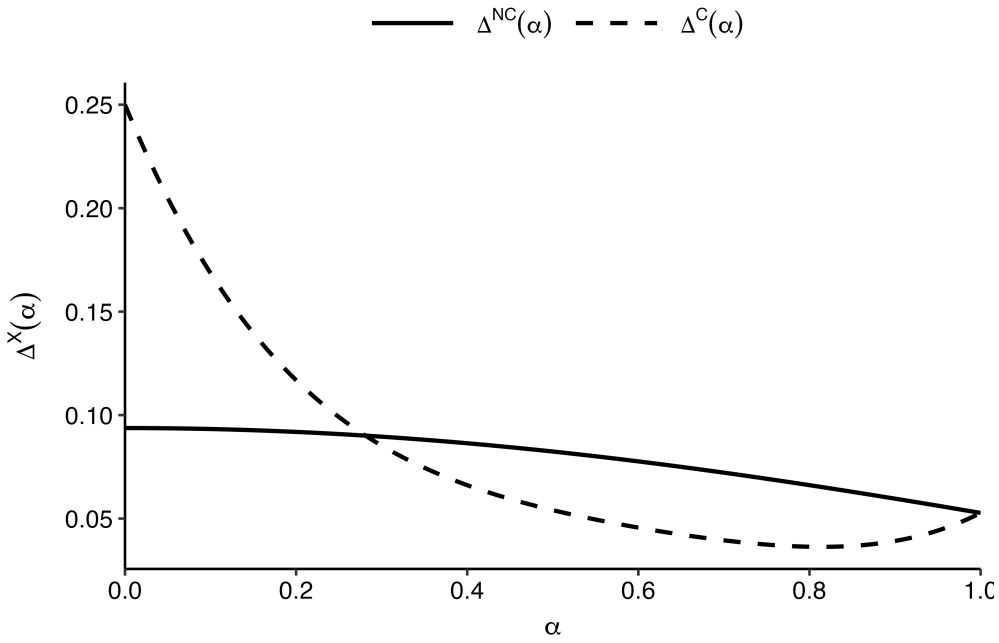


Figure 1: $\Delta^{NC}(\alpha)$ and $\Delta^{C}(\alpha)$ with $N = 5$ and $q = 0.75$.

This sign change is central for comparing equilibrium acquisition probabilities across the two games.

2.3 Equilibrium

In each game $X \in \{NC, C\}$, acquiring information is the unique best response if $\Delta^X(\alpha) > c$, not acquiring is the unique best response if $\Delta^X(\alpha) < c$, and any mixture between the two is a best response if $\Delta^X(\alpha) = c$. Accordingly, the equilibrium set in each game depends on the information cost c. The next lemma characterizes how the equilibrium set varies with c. Recall from Lemma 2 that $\Delta^C(\alpha) = \Delta^{NC}(\alpha)$ if and only if $\alpha = \hat{\alpha}$ or $\alpha = 1$. Define $\hat{c} := \Delta^{NC}(\hat{\alpha}) = \Delta^C(\hat{\alpha})$ and $\Delta(1) := \Delta^{NC}(1) = \Delta^C(1)$. For the remainder of this section, I fix any $n \geq 1$ and $q \in (0.5, 1)$ unless specified otherwise.

Lemma 3 (Equilibrium)**.** *The following hold:*

1. *If $c \geq \Delta^C(0)$, each game has the unique equilibrium $\alpha^{*NC} = \alpha^{*C} = 0$.*
2. *If $c \in (\Delta(1), \Delta^C(0))$, each game has a unique equilibrium, with $\alpha^{*C} \in (0, 1)$ and $\alpha^{*NC} \in [0, 1)$. Moreover, $\alpha^{*C} > \alpha^{*NC}$ if $c > \hat{c}$, and $\alpha^{*C} < \alpha^{*NC}$ if $c < \hat{c}$.*
3. *If $c \leq \Delta(1)$, game NC has the unique equilibrium $\alpha^{*NC} = 1$, whereas game C admits $\alpha = 1$ as an equilibrium and may also admit equilibria with $\alpha < 1$.*

communication, and enough uninformed-vote noise to offset that imbalance and preserve the value of her vote without communication. As α approaches one, the latter source disappears. At $\alpha = 1$, there are no uninformed voters, so both the persuasion and substitution effects vanish and $\Delta^C(1) = \Delta^{NC}(1)$.

When $c \geq \Delta^C(0)$, information is too expensive, and each game has the unique no-acquisition equilibrium. When $c \leq \Delta(1)$, information is sufficiently cheap that full acquisition is an equilibrium in both games, although game C may additionally admit equilibria with lower acquisition rates. The key comparison arises for intermediate costs, $\Delta(1) < c < \Delta^C(0)$, where each game has a unique equilibrium. At $c = \hat{c}$, both games have the common equilibrium acquisition rate $\hat{\alpha}$. For higher costs, equilibrium acquisition is below $\hat{\alpha}$, placing the equilibrium in the region where the persuasion effect dominates; communication therefore raises acquisition, so $\alpha^{*C} > \alpha^{*NC}$. For lower costs, equilibrium acquisition is above $\hat{\alpha}$, placing the equilibrium in the region where the substitution effect dominates; communication therefore crowds out acquisition, so $\alpha^{*C} < \alpha^{*NC}$.

When a game admits multiple equilibria, α^{*X} denotes an arbitrary equilibrium acquisition rate; all equilibrium comparisons in the remainder of this section hold for every such rate. The next proposition consolidates Lemma 3 into a single-cutoff comparison.

Proposition 1 (Information acquisition)**.** *There exists a unique cutoff $\hat{c} \in (\Delta(1), \Delta^C(0))$ such that $\alpha^{*C} \geq \alpha^{*NC}$ if $c \geq \hat{c}$ and $\alpha^{*C} \leq \alpha^{*NC}$ if $c \leq \hat{c}$. The inequality is strict for $c \in (\Delta(1), \Delta^C(0))$ with $c \neq \hat{c}$.*

Equilibrium acquisition under communication is weakly higher than under no communication when information is relatively expensive and weakly lower when it is relatively cheap, with strict inequality for intermediate costs except at $c = \hat{c}$. Put differently, when free riding is severe absent communication, communication mitigates the free-riding problem; when free riding is relatively mild, communication exacerbates it. This is the key result of the model and the main prediction tested in the experiment.

2.4 Group accuracy and welfare

I now turn to what these acquisition effects imply for the quality of group decisions and welfare. For each game $X \in \{NC, C\}$, let $G^X(\alpha)$ denote the ex ante probability that the group decision matches the true state when each voter acquires information with probability α, i.e., $G^X(\alpha) := \Pr(d = z \mid \alpha_i = \alpha \ \forall i \in N)$. I first show that, for any fixed acquisition rate, communication improves information aggregation:

$$G^C(\alpha) \geq G^{NC}(\alpha) \quad \text{for all } \alpha \in [0, 1], \tag{3}$$

with equality at $\alpha \in \{0, 1\}$ and strict inequality for $\alpha \in (0, 1)$. With communication, full revelation and sincere voting imply that the collective decision coincides with the Bayesian-optimal choice given the realized type profile. Without communication, the majority out-

come is a noisier function of the same type profile, because uninformed votes introduce noise. Hence, conditional on any realized type profile, the probability of a correct group decision is weakly higher in C than in NC.

Define equilibrium group accuracy as $G^{*X} := G^X(\alpha^{*X})$. The next proposition gives the equilibrium comparison, taking into account the effect of communication on acquisition established in Proposition 1.

Proposition 2 (Group accuracy)**.** *There exists a unique cutoff $\hat{c}_G \in (\Delta(1), \hat{c})$ such that $G^{*C} \geq G^{*NC}$ if $c \geq \hat{c}_G$ and $G^{*C} \leq G^{*NC}$ if $c \leq \hat{c}_G$. The inequality is strict for $c \in (\Delta(1), \Delta^C(0))$ with $c \neq \hat{c}_G$.*

The pattern in Proposition 2 mirrors that for information acquisition. Communication improves group accuracy when information is relatively expensive and reduces it when information is relatively cheap, with strict inequality for intermediate costs except at $c = \hat{c}_G$. The cutoff $\hat{c}_G$ at which the effect on accuracy flips is below the cutoff $\hat{c}$ at which the effect on acquisition flips. When $c > \hat{c}$, communication increases acquisition and also improves aggregation, yielding more accurate group decisions. When $\hat{c}_G < c < \hat{c}$, communication reduces acquisition but still improves accuracy, because its aggregation advantage outweighs the reduction in acquired information. When $c < \hat{c}_G$, that advantage is overwhelmed by amplified free riding, and accuracy declines under communication.

I next turn to welfare. For each game $X \in \{NC, C\}$, define welfare conditional on an acquisition rate α as $W^X(\alpha) := G^X(\alpha) - \alpha c$, i.e., the ex ante expected payoff from the group decision net of information costs.[9] Define equilibrium welfare as $W^{*X} := W^X(\alpha^{*X})$. The next proposition characterizes how communication affects equilibrium welfare.

Proposition 3 (Welfare)**.** *There exists a unique cutoff $\hat{c}_W \in \big(\Delta(1), \hat{c}_G\big)$ such that $W^{*C} \geq W^{*NC}$ if $c \geq \hat{c}_W$ and $W^{*C} \leq W^{*NC}$ if $c \leq \hat{c}_W$. The inequality is strict for $c \in \big(\Delta(1), \Delta^C(0)\big)$ with $c \neq \hat{c}_W$.*

The pattern again mirrors those for acquisition and accuracy, with strict inequality for intermediate costs except at $c = \hat{c}_W$. Figure 2 summarizes the three comparisons and the ordering $\hat{c}_W < \hat{c}_G < \hat{c}$. When $\hat{c}_W < c < \hat{c}_G$, communication reduces group accuracy but still increases welfare, because the savings in acquisition costs outweigh the loss in accuracy. When $c < \hat{c}_W$, information is cheap enough that this cost saving is small, and the comparison is dominated by the decline in accuracy induced by stronger free riding.

[9] As a social-optimum benchmark, I also show that equilibrium information acquisition is weakly below the welfare-maximizing level in both games, and strictly below it for intermediate information costs. Thus, communication can mitigate or exacerbate free riding, but the underprovision of information persists at intermediate costs, and it never generates over-acquisition relative to the welfare-maximizing benchmark.

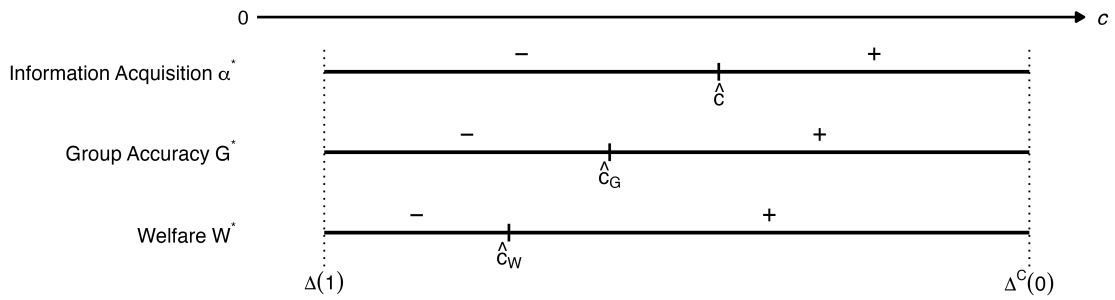


Figure 2: Thresholds for the effects of communication.

Notes: Minus (plus) signs indicate lower (higher) equilibrium outcomes under communication than under no communication. Cutoff locations are schematic.

Remarks on harmful deliberation. When $c \in (\Delta(1), \hat{c}_W)$, the unique equilibrium with communication yields strictly lower welfare than the equilibrium without communication.[10] This welfare loss highlights a tension between ex post information aggregation and ex ante incentives to acquire information. Conditional on the information acquired, communication implements the Bayesian-optimal collective decision, but ex ante this improvement can weaken acquisition incentives enough to reduce both group accuracy and welfare. Thus, making better use of information after it has been acquired need not improve welfare when the supply of information is endogenous. Two connections to the literature are worth noting.

First, this tension is related to the insights in Li (2001) and Gerardi and Yariv (2008). Li (2001) shows that distorting the collective decision rule away from the ex post optimal rule can strengthen incentives to acquire information and improve ex ante welfare, while Gerardi and Yariv (2008) show more generally that an ex ante optimal collective-choice mechanism need not be ex post efficient. Rather than solving for an optimal aggregation device, I compare two given institutions and show that allowing communication improves the ex post use of acquired information but can nevertheless lower ex ante welfare by exacerbating free riding in information acquisition.

Second, the result is related to, but distinct from, findings that communication *can be harmful* in collective decisions. In Gerardi and Yariv (2007) and Iaryczower et al. (2018), where information is taken as exogenously given, communication can support "bad" equilibria with lower group accuracy than without communication.[11] The mechanism here is different: welfare falls not because communication induces inefficient use of a fixed stock of information, but because more effective information sharing discourages information acquisi-

[10]This comparison is within the class of maximally revealing equilibria studied throughout; babbling equilibria remain outcome-equivalent to game NC.

[11]Specifically, these papers use the notion of communication equilibrium (Forges, 1986; Myerson, 1982), which permits correlated voting recommendations that can induce adverse inference.

tion in the first place. Moreover, in those papers the "bad" equilibria that perform worse than no communication coexist with equilibria that perform better, whereas for $c \in (\Delta(1), \hat{c}_W)$ the welfare loss in the present model arises in the unique equilibrium of game C within the class considered.

2.5 Discussion

Robustness. The two-sided effect of communication is not specific to the baseline model. Appendix C.1 relaxes several features one at a time in a three-voter environment and shows that the effect survives an unbalanced prior and informative but sufficiently less precise signals for non-acquirers, as well as asymmetric pure-strategy equilibria and imperfect information transmission. In each case, it arises for the same reason as in the baseline model: communication changes how strongly one voter's information affects the collective decision, and hence how strongly others' information substitutes for one's own.

Abstention. The model rules out abstention. Both effects of communication operate through the noise introduced by uninformed votes, and abstention offers another way to remove this noise. Appendix C.2 shows that game NC admits a swing-voter-curse continuation equilibrium in which non-acquirers abstain. Abstention then removes the uninformed-vote noise, just as communication does in the baseline model, and communication no longer affects acquisition incentives. Game NC also admits a full-participation continuation, however, in which uninformed voters randomize between the two alternatives and the benchmark comparisons go through unchanged. Whether communication matters for acquisition when abstention is allowed thus depends on whether non-acquirers abstain or participate.[12]

Large groups. Although this paper focuses on committees, the model also has implications for large electorates. Since $c \geq q - \frac{1}{2}$ rules out acquisition in both games at any group size, consider $c < q - \frac{1}{2}$. Appendix C.3 shows that, for any such c, once the group is sufficiently large, game NC has the unique equilibrium $\alpha^{*NC} = 0$, while game C has a unique equilibrium $\alpha^{*C} \in (0, 1)$, with $G^{*C} > \frac{1}{2} = G^{*NC}$ and $W^{*C} > \frac{1}{2} = W^{*NC}$. Thus, in the limit of large groups, communication helps regardless of the cost of information. Intuitively, in equilibrium, vote-pivotal events become negligible as the group grows, so the substitution effect vanishes, whereas the probability of information-pivotal events, and hence

[12]Experimental evidence is mixed: uninformed voters often abstain in standard swing-voter-curse experiments (Battaglini et al., 2010), but vote at substantial rates when information acquisition is endogenous (Großer and Seebauer, 2016; Elbittar et al., 2020).

the persuasion effect, remains bounded away from zero.[13] Communication even becomes necessary for sustaining *any* information acquisition, and hence for group decisions better than chance: without it, equilibrium acquisition collapses to zero. The two-sided effects in Propositions 1–3 are therefore most relevant for finite groups, such as committees and expert panels, rather than for large electorates.

3 Experimental Design

3.1 Parameterization and hypotheses

The experiment specializes the model to $N = 3$ and sets signal accuracy to $q = 0.9$. With three voters, the two opposing effects of communication characterized in Section 2 correspond to simple contingencies in the other members' acquisition decisions.[14] Consider a voter deciding whether to acquire information:

i) When neither other member acquires information, the persuasion effect arises: communication raises the value of acquisition. With communication, the voter's signal is shared with the group and determines the collective decision; without communication, it affects only her own vote and changes the collective decision only when the other two votes are split.

ii) When exactly one other member acquires information, the substitution effect arises: communication lowers the value of acquisition. With communication, the other member's signal is shared with the group, and acquiring a second equally precise signal does not increase expected decision accuracy. Without communication, the informed vote can be offset by the uninformed member's vote, so an additional signal retains positive value.

iii) When both other members acquire information, communication does not affect the value of an additional signal. In either game, the additional signal matters only when the other two signals disagree.

The experiment uses a 2×2 between-subject design that varies the availability of communication (NC versus C) and the cost of a signal (low, $c = 0.1$, versus high, $c = 0.25$). The two cost levels are chosen so that the model predicts opposite and substantial effects

[13] Relatedly, Hoshino and Gomberg (2025) show that, with rationally inattentive voters, opinion polls raise the probability that a large electorate chooses correctly; there, polls work by revealing how close the election is, rather than by sharing acquired information.

[14] With more voters, which effect arises also depends on the realized signals.

of communication on equilibrium information acquisition. Table 1 reports the symmetric equilibrium acquisition rates.[15]

Table 1: Symmetric Equilibrium Information Acquisition Rates

	Low cost ($c = 0.10$)	High cost ($c = 0.25$)
No communication (NC)	0.884	0
Communication (C)	0.560	0.215

At $c = 0.1$, acquisition is relatively high without communication: the predicted rate is 0.884 in game NC. This makes the contingency in which exactly one of the other two members acquires much more likely than the contingency in which neither acquires. The substitution effect therefore dominates, and communication lowers the predicted acquisition rate to 0.560. At $c = 0.25$, by contrast, free riding is so severe without communication that the predicted acquisition rate is zero. Thus, neither other member acquires in the NC equilibrium, so the persuasion rather than the substitution effect is operative when communication is introduced. Communication then sustains a positive equilibrium acquisition rate of 0.215. These predictions yield the two main hypotheses.[16]

H1 (Acquisition at low cost). *When the information cost is low* ($c = 0.1$)*, communication reduces information acquisition.*

H2 (Acquisition at high cost). *When the information cost is high* ($c = 0.25$)*, communication increases information acquisition.*

The model also generates predictions for group outcomes under the same parameterization. Communication reduces both group accuracy and welfare at the low information cost, whereas it increases both at the high information cost. These comparisons reflect both communication's effect on equilibrium information acquisition and its direct effect on information aggregation. The corresponding equilibrium values are reported in Appendix D.1.

H3 (Group outcomes at low cost). *When the information cost is low* ($c = 0.1$)*, communication reduces group accuracy and welfare.*

H4 (Group outcomes at high cost). *When the information cost is high* ($c = 0.25$)*, communication increases group accuracy and welfare.*

[15]For comparison, under the experimental parameterization, the welfare-maximizing symmetric acquisition rates are 1 in both games at the low cost, and 0.955 in NC and 0.630 in C at the high cost. Equilibrium acquisition thus lies below the welfare-maximizing level in all four treatments.

[16]The directional predictions for information acquisition are unchanged under asymmetric pure-strategy equilibria: under the experimental parameterization, communication reduces the number of acquirers from two to one at the low cost and increases it from zero to one at the high cost, as shown in Appendix C.1.3.

3.2 Implementation and procedures

Experimental task. The experiment implements games C and NC, which differ only in the availability of a communication stage before voting. In each round, the computer selects one of two jars, labeled Red and Blue, with equal probability. Subjects do not observe which jar was selected but know that both are equally likely. Each member receives a reward of 1,000 points if the group decision matches the selected jar and no reward otherwise, before any information-acquisition costs. The Red jar contains nine red balls and one blue ball, and the Blue jar nine blue balls and one red ball, so the color of a ball drawn from the selected jar provides a signal that identifies the selected jar correctly with probability $q = 0.9$. Each round then proceeds through the following decision stages.

Information acquisition. Each subject decides whether to pay the treatment-specific cost of 100 or 250 points to draw a ball from the selected jar. If he or she draws, the computer randomly picks one of the ten balls in the selected jar and displays its color to him or her alone, and the cost is deducted when the round payoff is computed. If he or she does not draw, no cost is deducted. Balls are drawn independently across group members. Subjects are not informed whether the other members drew.

Communication. In the communication treatments, each subject then chooses one of three options: "I have drawn a red ball," "I have drawn a blue ball," or no message. All three options are available regardless of whether the subject drew a ball or what color was drawn. While the theory allows for unrestricted communication, the experiment uses this finite message space to keep the message data structured and easily interpretable. The restriction preserves the cheap-talk nature of the stage and is rich enough to permit full revelation, leaving the equilibrium predictions studied here unchanged. Once all members have submitted their messages, the messages are revealed to the group. The no-communication treatments omit this stage.

Voting. Each subject then votes for either Red or Blue. Abstention is not allowed. Once all members have submitted their votes, the group's decision is determined by majority rule, and the round ends.

At the end of the round, the computer displays the true jar, the group decision, and the vote count for each jar, but does not disclose the other group members' acquisition choices. Each subject also sees his or her own earnings for the round, which are not shown to others. At any point, subjects can click "Show History" to review their own information and decisions, along with the public information from their previous rounds.

Additional measures. Subjects completed 30 rounds of the experimental task described above. During and immediately after the final round, I collected three additional measures related to acquisition behavior, each elicited once and unincentivized.[17] These measures are used for the diagnostic analyses in Section 5.

Beliefs about others' acquisition. Before making the acquisition decision in round 30, subjects reported their beliefs about the acquisition behavior of the other two members of their group in that round, allocating probabilities across the events that zero, one, or both of them draw a ball.

Perceived value of acquisition (PVA). On the same screen and after the beliefs questions, subjects then reported the probability that their group would decide correctly in that round if they drew a ball and if they did not. I define the perceived value of acquisition as the difference between these two probabilities, i.e., the perceived change in the probability of a correct group decision from acquiring information. Because the reward for a correct decision is normalized to one, PVA is also the perceived expected-payoff gain from acquisition to be weighed against the normalized cost c. Round 30 was then played as usual and remained eligible for payment.

Conditional choices. Immediately after the final round, subjects answered three hypothetical questions asking whether they would draw a ball if none, one, or both of the other two group members drew, choosing among "Yes," "No," and "Not sure." These hypothetical choices remove uncertainty about others' acquisition behavior and isolate subjects' responses to the contingencies underlying the persuasion and substitution effects.

Procedures. At the start of each session, subjects were randomly assigned to matching groups of six, which remained fixed throughout the session, and each matching group was assigned to a single treatment. Subjects then read the instructions for their treatment on screen, completed a comprehension quiz that had to be answered correctly to proceed, and played one unpaid practice round before the formal rounds began. In every round, subjects were randomly rematched into groups of three within their matching group. A jar was independently selected for each group in every round.

One of the 30 rounds was randomly selected for payment. Experimental points were converted at a rate of 100 points per euro. Subjects additionally received a base payment of

[17] These measures were elicited only at the end of the experiment to minimize potential demand effects from prompting subjects to think explicitly about aspects of the acquisition decision. One of these measures involves asking about the probability of a correct group decision conditional on drawing and on not drawing. Because only one acquisition decision is realized, the two assessments cannot both be incentivized. For consistency and to keep the payment scheme simple, the other two measures were also elicited without incentives. Subjects were told that these answers would not affect their earnings and were asked to respond as accurately as possible.

€6.50. Because a subject could incur the high information-acquisition cost and receive no reward from the group decision, earnings in the selected round could be as low as −€2.50. The base payment therefore guaranteed a minimum total payment of €4, the show-up fee announced in the recruitment materials.

After the main task, subjects completed three short incentivized tasks eliciting individual characteristics: risk attitude (Gneezy and Potters, 1997), attitude toward strategic uncertainty (Heinemann et al., 2009; Alós-Ferrer and Garagnani, 2020), and social preferences measured using Social Value Orientation (SVO) (Murphy et al., 2011). One of the three tasks was randomly selected for payment, with the resulting earnings added to the base payment and earnings from the main task. Subjects then completed a demographic questionnaire. Detailed descriptions of these tasks and measures, together with balance checks and supplementary analyses using them, are reported in Appendix D.2.

The experiment was conducted at the CentERlab of Tilburg University between May and August 2025. The experimental design, H1 and H2, and the main analyses were pre-registered.[18] In total, 288 subjects participated, earning €14.51 on average and spending about 40 minutes. One high-cost no-communication matching group is excluded because of a procedural error.[19] The resulting analysis sample contains 282 subjects in 47 matching groups. The matching group is the unit of independent observation: unless otherwise noted, nonparametric tests are based on matching-group averages, and regression standard errors are clustered at the matching-group level.

4 Experimental Results

4.1 Information acquisition

Figure 3 reports information acquisition rates by treatment.[20] Acquisition is higher under communication at both cost levels. At low cost ($c = 0.1$), the average acquisition rate is 0.520 without communication and 0.618 with communication. The difference runs counter

[18] The pre-registration is available at https://aspredicted.org/r62as4.pdf.

[19] One subject initially sat at a seat assigned to a different treatment and read that treatment's instructions before the mistake was noticed and corrected. Because subjects were repeatedly rematched within fixed matching groups, I exclude the entire matching group rather than only the affected subject. This exclusion was not specified in the pre-registration; Appendix D.3 reports results for the full sample.

[20] Observed acquisition rates are generally more interior than the symmetric equilibrium predictions. At the low cost, acquisition is below its prediction without communication (0.520 versus 0.883), while at the high cost it is above its prediction both without communication (0.308 versus 0) and with communication (0.384 versus 0.215). With communication at the low cost, acquisition is relatively close to its prediction (0.618 versus 0.560). Because the hypotheses concern the direction of communication effects rather than treatment-specific levels, I report tests and further analysis of these departures in Appendix E.2, where quantal-response fits substantially narrow the gap between predicted and observed levels.

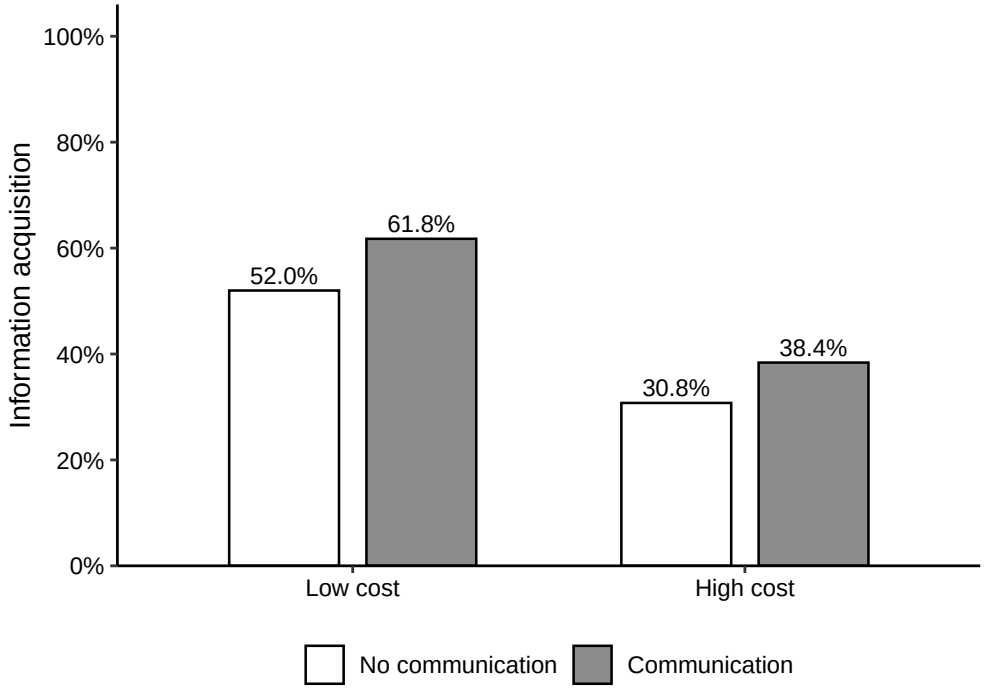


Figure 3: Information acquisition by treatment

to H1, which predicts a decrease in acquisition under communication, and is not statistically significant (two-sided Mann–Whitney test, $p = 0.11$).[21] At high cost ($c = 0.25$), the average acquisition rate is 0.308 without communication and 0.384 with communication. The difference is in the direction predicted by H2 and is statistically significant (one-sided Mann–Whitney test, $p = 0.042$).[22]

Table 2 reports OLS regressions of the acquisition decision. Pooling across cost levels, communication significantly increases acquisition by 8.7 percentage points (Column (1)). Allowing the communication effect to differ by cost yields increases of 9.8 percentage points at the low cost and 7.6 percentage points at the high cost (Column (3)), both significant at the 10% level. The interaction between communication and high cost is small and not statistically significant. Thus, the regression results are consistent with the pattern in Figure 3: the communication effect is positive at both cost levels, with no evidence that it differs across cost levels. Columns (2) and (4) examine how acquisition changes over the course of the experiment. Acquisition declines modestly over rounds, while the gap between the communication and no-communication treatments widens slightly. The positive effect

[21]Since H1 and H2 were preregistered as directional hypotheses, I report one-sided tests in the predicted direction, and a two-sided test where the observed difference has the opposite sign.

[22]Studentized permutation Brunner–Munzel tests, which do not require identical distributions under the null, yield the same conclusions (low cost: two-sided $p = 0.101$; high cost: one-sided $p = 0.040$). Retaining the matching group excluded for procedural reasons yields $p = 0.102$ in both the one-sided Mann–Whitney and permutation Brunner–Munzel tests at the high cost.

Table 2: Effect of communication on information acquisition

	(1)	(2)	(3)	(4)
Communication	0.087^{**}	0.087^{**}	0.098^{*}	0.098^{*}
	(0.035)	(0.035)	(0.052)	(0.052)
High cost	-0.223^{***}	-0.223^{***}	-0.212^{***}	-0.212^{***}
	(0.034)	(0.034)	(0.055)	(0.055)
Communication × High cost			−0.021	−0.021
			(0.069)	(0.069)
Round (centred)		-0.003^{***}		-0.003^{***}
		(0.001)		(0.001)
Round × Communication		0.002^{*}		0.002^{*}
		(0.001)		(0.001)
Comm. effect at high cost			0.076^{*}	0.076^{*}
			(0.045)	(0.045)
one-sided p			0.050	0.050
Observations	8,460	8,460	8,460	8,460
R^2	0.057	0.059	0.057	0.059
Matching groups	47	47	47	47

Notes: OLS; the dependent variable is an indicator for acquiring a signal. Standard errors are clustered at the matching-group level. Columns (1) and (2) pool the communication effect across cost levels; Columns (3) and (4) allow it to differ by cost. In Columns (3) and (4), the coefficient on Communication gives the effect at the low cost, and the reported high-cost effect is $\beta_{\text{Comm}} + \beta_{\text{Comm}\times\text{High}}$, for which the one-sided p-value for H2 is reported separately. Stars are based on two-sided tests throughout: $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

of communication therefore does not dissipate with experience.[23] Moreover, across specifications, acquisition is substantially less likely when the information cost is high, in line with previous experimental evidence that information acquisition is responsive to its cost (Bhattacharya et al., 2017).

Result 1 (Information acquisition)**.** *Communication increases information acquisition at the high cost, as predicted by H2, but does not reduce it at the low cost, contrary to H1. The effect is significantly positive when pooled and does not vary significantly by cost.*

4.2 Communication and Voting

The theoretical analysis focuses on equilibria with full revelation and sincere voting. More generally, communication can affect information aggregation and acquisition only insofar as it transmits information and votes respond to the information available at the voting stage. Both features are strongly present in the experiment.

[23] Appendix D.4 reports the round-by-round acquisition pattern, the distribution of individual acquisition frequencies, and additional robustness checks.

Communication was highly informative, though not fully revealing. Subjects who acquired information almost always reported their realized signal truthfully: they did so in 99.9% of cases at the low cost and 98.7% at the high cost. Subjects who did not acquire information nevertheless sent a message claiming to have observed a signal in 16.1% of cases at the low cost and 10.1% at the high cost. Despite these false claims, signal messages remained strongly informative about the state: the alternative indicated by a signal message was the true state in 86.4% of cases at the low cost and 84.2% at the high cost,[24] compared with benchmarks of 90% under full revelation and 50% under uninformative (babbling) communication.

Voting behavior conformed closely to sincere voting based on Bayesian posteriors. Table 3 reports the frequency of voting for Red conditional on a subject's own signal and the messages sent by the other two group members. In the communication treatments, consider subjects who acquired a red signal (the pattern for a blue signal is nearly symmetric). When the available information favored Red—under message profiles (R, B), $(-, -)$, $(R, -)$, and (R, R)—subjects voted for Red almost always. When both other members sent blue messages, (B, B), their two concordant messages outweighed the subject's red signal,[25] and subjects voted for Blue in 97% of cases. When one other member sent a conflicting blue message and the remaining member sent no message, $(B, -)$, subjects followed their own red signal in 82% of cases.[26] Uninformed subjects similarly responded strongly to messages: they almost always followed a single or two concordant signal messages, while splitting approximately evenly when the two messages conflicted or when neither member sent a signal message. In the no-communication treatments, subjects who acquired information voted according to their own signal in 99.7% of cases, while uninformed subjects split approximately evenly between the two alternatives.

4.3 Group Accuracy and Welfare

I next examine group accuracy, defined as the proportion of correct group decisions, and average payoff, defined as the average normalized round payoff across group members and

[24] These percentages are calculated as the observed proportion of signal messages that match the true state. This proportion is higher at the low cost despite the higher rate of false claims among non-acquirers because a larger share of signal messages was backed by actual acquisition.

[25] Given one's own signal of precision $q = 0.9$, two concordant opposing messages yield a posterior favoring their alternative whenever message accuracy exceeds 0.75, which observed message accuracy exceeded at both cost levels.

[26] This is also consistent with sincere voting. Under full revelation, the two conflicting signals cancel out, leaving a posterior of 1/2, so sincere voting does not pin down the vote. Under imperfect revelation, when message accuracy is below signal precision, as in the data, the posterior favors one's own signal, so following it is sincere.

Table 3: Frequency of Voting for Red

	C: others' messages						NC
Own signal	B,B	B,–	R,B	–,–	R,–	R,R	
–	0.02 (236)	0.05 (529)	0.53 (152)	0.54 (447)	0.94 (539)	0.98 (254)	0.50 (2408)
R	0.03 (30)	0.82 (102)	1.00 (98)	0.99 (205)	1.00 (421)	1.00 (229)	0.99 (917)
B	0.00 (256)	0.00 (373)	0.00 (98)	0.01 (213)	0.21 (100)	0.97 (38)	0.00 (815)

Note: The table pools cost levels because voting patterns are similar; cost-specific results are reported in Appendix D.5. R and B denote red and blue for own signals and others' messages; "–" denotes no acquisition (rows) or no message (columns). Columns are ordered by weakly increasing net support for Red. Entries report the fraction voting for Red, with observation counts in parentheses.

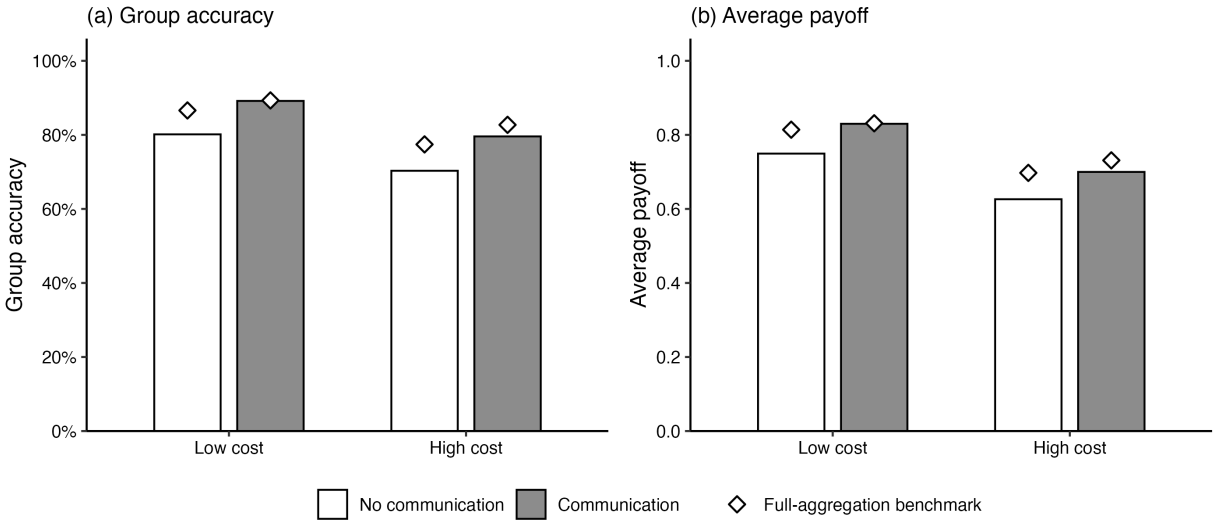


Figure 4: Group accuracy and average payoff by treatment.

used as a measure of social welfare. Figure 4 reports both outcomes across treatments. Bars show observed outcomes, while markers show the full-aggregation benchmarks defined below. Communication improves group accuracy at both cost levels. At the low cost, the fraction of correct group decisions increases from 80.1% without communication to 89.2% with communication (two-sided Mann–Whitney test, $p = 0.004$). At the high cost, it increases from 70.3% to 79.6% ($p = 0.015$). Average payoffs likewise increase with communication, from 0.749 to 0.830 at the low cost ($p = 0.003$) and from 0.626 to 0.700 at the high cost ($p = 0.029$). As with information acquisition, the predicted benefit of communication at the high cost is borne out, but its predicted downside at the low cost does not materialize.

Result 2 (Group outcomes)**.** *Communication increases group accuracy and average payoffs at both cost levels, consistent with H4 but contrary to H3.*

The increase in group accuracy can reflect both changes in information acquisition and improved aggregation of the information acquired. To distinguish these channels, I construct a *full-aggregation benchmark* for group accuracy: the accuracy that would obtain if a single

decision maker observed all signals acquired in the group and chose the alternative optimally.[27] I define the gap between this benchmark and observed accuracy as the *aggregation shortfall*. Treatment effects on the benchmark capture acquisition-induced changes in attainable accuracy, whereas effects on the shortfall capture changes in aggregation efficiency. At the low cost, communication raises the benchmark only modestly, from 86.6% to 89.3% (group-round-level OLS, $p = 0.195$), but nearly eliminates the aggregation shortfall: without communication, observed accuracy falls 6.5 percentage points below the benchmark, whereas with communication it essentially reaches the benchmark ($p < 0.001$ for the difference in shortfalls). Thus, the low-cost accuracy gain is driven primarily by improved aggregation, while the predicted acquisition-induced loss in attainable accuracy does not materialize. At the high cost, both channels contribute: communication raises the benchmark by 5.3 percentage points and reduces the shortfall by 4.0 percentage points, although both estimates are less precise ($p = 0.099$ and $p = 0.114$). Appendix D.6 reports the full decomposition, including for welfare, and alternative decompositions.

5 Understanding the missing substitution effect

The key experimental finding is that communication increases information acquisition and improves collective outcomes, with no evidence of its predicted harmful effect at the low cost. This section investigates why. Rather than trying to explain every quantitative departure from the equilibrium predictions, I focus on this qualitative puzzle: the persuasion effect appears operative in the data, whereas the predicted substitution effect is missing.

Several candidate explanations appear to have limited explanatory power for this asymmetry. One is post-acquisition equilibrium selection. This does not seem to be the main driver, given the highly informative communication and broadly sincere voting observed in the data; moreover, moving to a less informative communication equilibrium would attenuate both the persuasion and substitution forces rather than naturally generate the observed asymmetry. Other candidates include departures from the assumed preferences, such as risk aversion or social preferences, as well as behavioral models such as level-k reasoning and quantal response. Appendix E examines what these alternatives can account for in the data and where they fail; it also discusses the quantitative departures from the equilibrium predictions.

[27] Given equally likely states and equally precise independent signals, the optimal decision follows the majority of the acquired signals; ties and rounds in which no signal is acquired are assigned an accuracy of one half.

5.1 Diagnostic evidence

I use three additional measures collected in the experiment—perceived value of acquisition, beliefs about others' acquisition, and conditional choices—to diagnose the source of the missing substitution effect.

Perceived value of acquisition. I first examine subjects' *perceived value of acquisition* (PVA). As described in Section 3, before making their acquisition decision in the final round, subjects reported how likely their group was to decide correctly if they drew a ball and if they did not. PVA is the difference between these two probabilities, and captures the perceived gain in group accuracy from acquisition. It is closely related to the model's (conditional) value of acquisition $\Delta^X(\alpha)$, but differs in that $\Delta^X(\alpha)$ is defined given that each other voter acquires information with probability α, whereas PVA is not conditioned on others' acquisition choices and therefore incorporates subjects' beliefs about others' acquisition behavior.

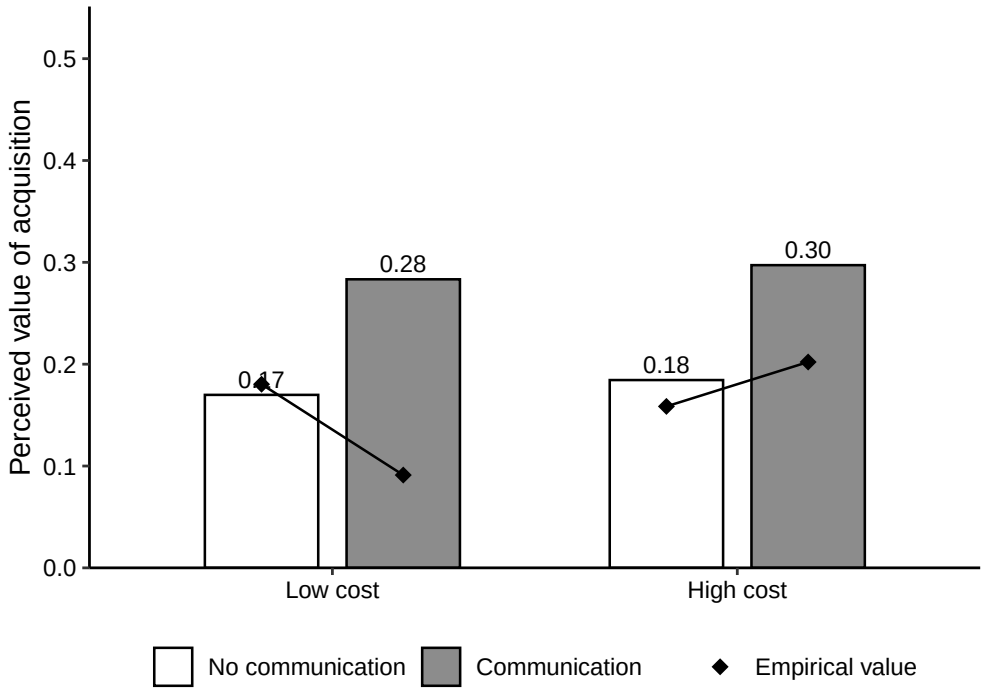


Figure 5: Perceived and empirical value of acquisition

Figure 5 displays PVA by treatment, together with an empirical counterpart computed from observed behavior. Communication substantially raises PVA at both information costs. At the low cost, mean PVA increases from 0.17 without communication to 0.28 with communication (two-sided Mann–Whitney test, $p = 0.006$); at the high cost, it increases from 0.18 to 0.30 ($p = 0.006$). Thus, communication raises the perceived accuracy gain from acquiring a signal by about 11 percentage points at either cost. This pattern is consistent with

the persuasion effect: subjects perceive communication as increasing the return to acquiring information.

The diamonds in Figure 5 report the empirical value of acquisition. For each matching group, I compute the change in average group accuracy from an additional acquisition conditional on facing zero, one, or two other acquirers, weighted by the frequency of each contingency. Two patterns stand out. First, while perceived values are statistically indistinguishable from empirical values without communication at either cost (Wilcoxon signed-rank tests: low cost, $p = 0.556$; high cost, $p = 0.307$), they substantially exceed their empirical counterparts under communication (low cost, $p = 0.003$; high cost, $p = 0.009$). Second, the discrepancy is especially stark at the low cost, where communication moves perceived and empirical values in opposite directions. At the high cost, the empirical value rises from 0.16 to 0.20 (two-sided Mann–Whitney test, $p = 0.045$), in the same direction as PVA but by much less. At the low cost, however, the empirical value actually falls from 0.18 to 0.09 ($p = 0.004$) when communication is available, consistent with the substitution effect, while PVA rises sharply.

Table 4: Information acquisition and perceived value of acquisition

	(1)	(2)
Communication	0.098*	0.021
	(0.052)	(0.047)
Perceived value of acquisition		0.678***
		(0.077)
High cost and × Communication	Yes	Yes
Round and × Communication	Yes	Yes
Comm. effect at high cost	0.076*	0.000
	(0.045)	(0.047)
one-sided p	0.050	0.502
Observations	8,460	8,460
R^2	0.059	0.145
Matching groups	47	47

Notes: OLS regressions of information acquisition. Standard errors are clustered at the matching-group level. Communication gives the effect at the low cost; the reported high-cost effect is the corresponding linear combination, and its p-value is one-sided in the direction of H2. Stars are based on two-sided tests: $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

PVA is also strongly associated with acquisition behavior, and its inclusion absorbs much of the estimated communication effect. Column (1) of Table 4 reproduces the last specification from Table 2, which allows the communication effect to vary by cost level and round, and Column (2) adds PVA. The coefficient on PVA is 0.678 and highly significant. Once PVA is included, the coefficient on communication falls from 0.098 to 0.021, and the

estimated communication effect at the high cost falls from 0.076 to nearly zero. Because PVA was elicited only once near the end of the experiment, I do not interpret these estimates as causal mediation. They nevertheless indicate that the communication effect operates largely through perceived returns rather than through some other channel.

Taken together, these results show that the missing substitution effect is visible in subjects' perceived economic incentives: subjects overvalue acquisition when communication is available, and at the low cost, communication raises the perceived value of acquisition even as its empirical value falls.[28] The PVA evidence alone, however, does not reveal whether this misperception comes from incorrect beliefs about others' acquisition behavior or from how subjects evaluate the return to acquisition conditional on others' choices. I next separate these two components. In what follows, I focus on the low-cost treatments, where the missing substitution effect creates the qualitative puzzle; Appendix D.7 reports the qualitatively similar high-cost results.

Beliefs about others' acquisition. One possible source of the overvaluation documented above is that subjects misperceive how much information others acquire. In particular, if subjects underestimate others' acquisition, they may assign too much value to acquiring an additional signal themselves. At the low cost, however, elicited beliefs are broadly consistent with the observed distribution of others' acquisition; Appendix D.7 reports the elicited probabilities and observed frequencies for zero, one, and two other acquirers.

More directly, I recompute the empirical value of acquisition using subjects' elicited beliefs rather than the observed frequencies. The resulting belief-weighted value is on average 0.114 with communication and 0.175 without communication. Thus, even given subjects' own beliefs, correctly assessing the return to acquisition conditional on others' choices would still imply a lower value under communication, consistent with the substitution effect and in sharp contrast to the observed PVA. The reweighting also closes only a small part of the gap between empirical and perceived values in the low-cost communication treatment. Belief errors about others' acquisition behavior therefore explain little of the overvaluation and do not appear to account for the missing substitution effect.

Conditional choices. I next examine subjects' reported responses across the contingencies. With three-member groups, theory predicts that communication raises the value of acquisition when neither other member acquires (the persuasion effect), lowers it when ex-

[28] This overvaluation neither generates excessive information acquisition nor fully eliminates the underprovision of information. Observed acquisition remains below the welfare-maximizing symmetric benchmark in all four treatments (see footnote 15); insofar as the overvaluation raises acquisition under communication, it moves acquisition toward, but still short of, that benchmark.

actly one other member acquires (the substitution effect), and leaves it unchanged when both acquire. I first document the realized conditional values in the experimental data and then turn to subjects' reported choices.

Panel (a) of Figure 6 reports the empirical value of acquisition separately when zero, one, or two other members acquire at the low cost. When neither other member acquires, communication raises the empirical value from 0.138 to 0.452 ($p = 0.001$); when exactly one other member acquires, it instead lowers the value from 0.233 to 0.006 ($p < 0.001$). When both other members acquire, the values are nearly identical, at 0.063 without communication and 0.059 with communication ($p = 0.921$). Thus, the realized conditional values closely reproduce the theoretical pattern. In particular, communication sharply reduces the value of an additional signal when exactly one other member acquires, consistent with the substitution effect. This conditional pattern is also consistent with the decline in the unconditional empirical value under communication documented above.

Panel (b) displays subjects' reported conditional acquisition choices. As described in Section 3, after completing all 30 rounds, subjects answered whether they would draw a ball if zero, one, or two other group members drew, choosing among "Yes," "No," and "Not sure" in each contingency. The figure reports the fraction choosing "Yes". Because these choices were hypothetical and unincentivized, I interpret them as diagnostic evidence rather than as additional incentivized treatment outcomes.[29]

Subjects' reported choices respond strongly to the persuasion incentive but not to the substitution incentive. When neither other member acquires, communication sharply raises the empirical value of acquisition, and the fraction stating that they would acquire rises from 54.2% without communication to 88.9% with communication (Mann–Whitney test, $p < 0.001$). When exactly one other member acquires, however, communication lowers the empirical value to essentially zero, yet stated acquisition is virtually unchanged: 61.1% without communication and 59.7% with communication ($p = 0.976$). Taken together, the evidence suggests that the missing substitution effect lies primarily in subjects' conditional responses rather than in mistaken beliefs about others' acquisition. Subjects appear responsive when communication increases the return to their own information, but much less responsive when it reduces that return. This asymmetric response is consistent with the PVA finding that subjects overvalue acquisition under communication.

[29]The post-experimental responses are nevertheless strongly associated with subjects' earlier incentivized behavior. In the low-cost communication treatment, the 36.1% of subjects reporting the theory-consistent pattern $(1, 0, 0)$ acquired information in 34.7% of the preceding rounds, compared with 77.0% among the remaining subjects, a difference of 42.3 percentage points (matching-group clustered s.e. = 7.9 percentage points, $p < 0.001$).

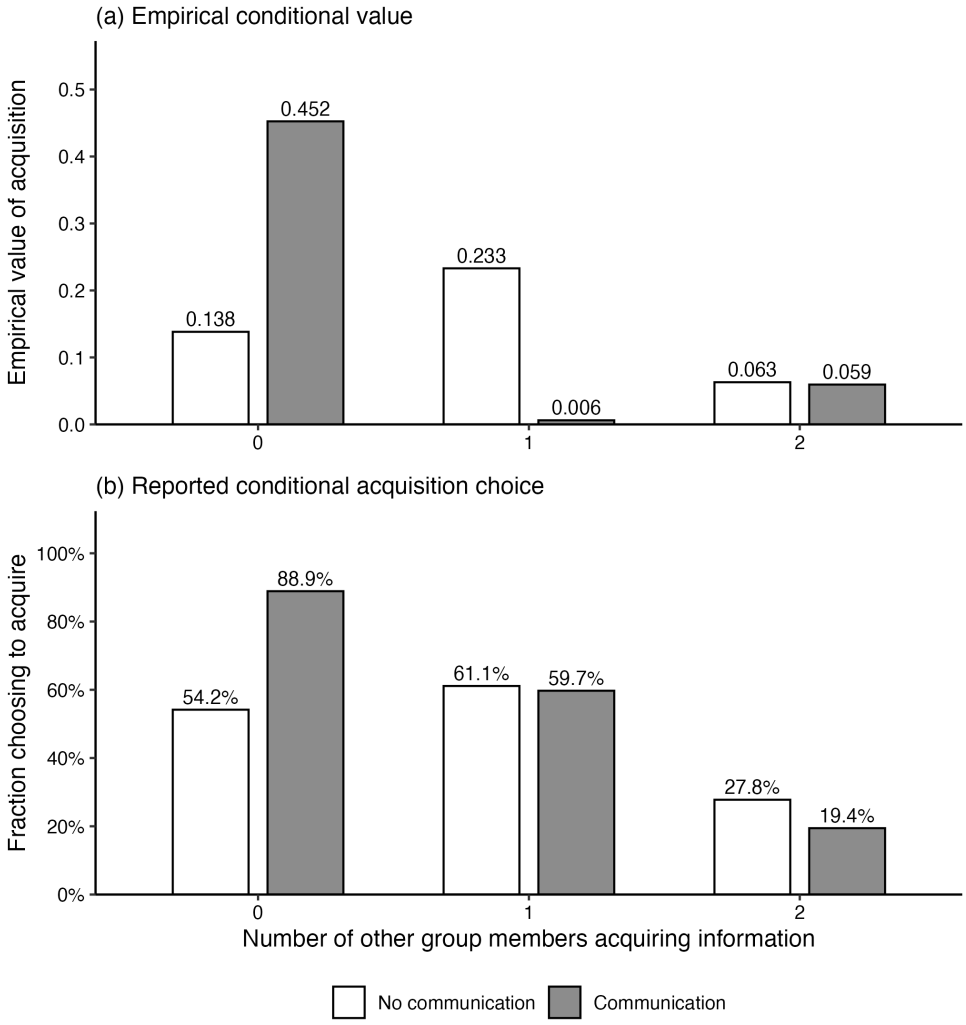


Figure 6: Conditional value of acquisition and reported acquisition choices at low cost.

5.2 Possible behavioral interpretations

The evidence above points to a specific failure when exactly one other member is informed: communication substantially lowers the value of acquisition, yet subjects' stated acquisition choices are almost unchanged. Why do subjects fail to respond to this substitution incentive? I consider two possible behavioral interpretations.

Under-inference from communication. One possibility is that subjects do not fully account for the dependence of an informed member's message on his or her private signal. As a result, they perceive others' messages as less informative than they actually are, making an additional signal of their own appear more valuable. This idea is closely related to cursed equilibrium and its sequential extension (Eyster and Rabin, 2005; Fong et al., 2025), in which a player partially neglects the dependence of another player's action on his or her private information.

I focus on a stage-specific version in which cursedness applies only to message interpretation, thereby isolating the inference failure suggested by the conditional-choice evidence. Let $\chi \in [0,1]$ parameterize the extent of cursed inference. Take truthful communication as the benchmark.[30] A voter perceives an informed member's message as having precision

$$\tilde{q}(\chi) = (1-\chi)q + \chi\frac{1}{2},$$

where $\chi = 0$ corresponds to correct inference and $\chi = 1$ to treating the message as uninformative about the sender's signal.[31]

When neither other member acquires, if the voter acquires, her message remains informative for any $\chi < 1$. The two uninformed voters therefore follow it, and the group is correct with probability q. Without acquisition, the group is correct with probability $1/2$. The perceived value of acquisition under communication is therefore $\widetilde{\Delta}_0^C(\chi) = q - 1/2$, the same as under correct inference. Thus, under-inference from communication does not change the perceived value of acquisition when neither other member acquires.

When exactly one other member acquires, the implication is different. If the voter does not acquire, the group follows the sole informed member's message, which the voter perceives as correct with probability $\tilde{q}(\chi)$. If the voter acquires, perceived group accuracy is q.[32] Therefore, the perceived value of acquisition under communication is

$$\widetilde{\Delta}_1^C(\chi) = q - \tilde{q}(\chi) = \chi\left(q - \frac{1}{2}\right).$$

Under-inference from communication therefore makes one's own acquisition appear more valuable when one other member is informed.

Because this distortion operates only through communication, it does not affect perceived acquisition values in game NC. When zero or one other member acquires, $\Delta_0^{NC} = \Delta_1^{NC} = \frac{1}{2}(q - \frac{1}{2})$. Hence, the persuasion effect, $\widetilde{\Delta}_0^C(\chi) - \Delta_0^{NC} = \frac{1}{2}(q - \frac{1}{2})$, is unchanged for

[30] Allowing for imperfect reporting changes the objective informativeness of messages but not the qualitative logic of the argument.

[31] Formally, conditional on acquisition, cursed inference replaces the signal-dependent message strategy $\sigma(m \mid s)$ by $\sigma^\chi(m \mid s) = (1-\chi)\sigma(m \mid s) + \chi\bar{\sigma}(m)$, where $\bar{\sigma}(m)$ is its type-averaged counterpart. Under truthful communication and ex ante symmetry, this gives $\tilde{q}(\chi) = (1-\chi)q + \chi/2$.

[32] For $\chi > 0$, an informed voter puts more weight on her own signal than on a conflicting message, while an uninformed voter follows agreeing messages and randomizes when they conflict. Hence, when two members acquire signals, if both signals are correct (with probability q^2), the group is correct; if both signals are incorrect (with probability $(1-q)^2$), the group is incorrect; and if the two signals conflict (with probability $2q(1-q)$), the two informed voters cast opposite votes and, by symmetry, the group is correct with probability $1/2$. Perceived group accuracy is therefore $q^2 + 2q(1-q)\frac{1}{2} = q$. The same value obtains at $\chi = 0$.

any $\chi < 1$. By contrast, when exactly one other member acquires,

$$\widetilde{\Delta}_1^C(\chi) - \Delta_1^{NC} = \left(\chi - \frac{1}{2}\right)\left(q - \frac{1}{2}\right).$$

The substitution effect is therefore attenuated as χ increases and disappears at $\chi = 1/2$.[33] This mirrors the conditional-choice pattern: communication strongly increases reported acquisition when neither other member acquires, but has essentially no effect when exactly one other member acquires. Appendix E reports a structural fit of the cursed-message-inference specification to the four treatment-level acquisition rates.

Overvaluing an additional noisy signal. A different possibility is that subjects overestimate the marginal decision value of an additional noisy signal. When exactly one other member acquires a signal, even a subject who fully appreciates its informativeness must recognize a subtle point to correctly evaluate the marginal value of her own acquisition: once one signal of precision q is already available to the group, a second independent signal of the same precision provides additional statistical information but does not increase the probability of choosing the correct alternative. With one signal, decision accuracy is q; with two signals, both are correct with probability q^2, while conflicting signals occur with probability $2q(1-q)$ and leave the group indifferent, so that either alternative is chosen with equal probability. Decision accuracy is therefore

$$q^2 + 2q(1-q)\cdot\tfrac{1}{2} = q.$$

The second signal is thus statistically informative—for example, agreement makes the posterior more extreme—but has zero incremental value for ex ante decision accuracy.[34]

Subjects may not fully distinguish these two notions of informational value. In particular, it may be natural to view another independent noisy signal as useful simply because the first signal is imperfect, even though its marginal decision value is zero in this environment. Related experimental evidence in individual decision making is consistent with

[33] With $q = 0.9$, $\chi = 1/2$ corresponds to a perceived message precision of 0.7, compared with the objective precision 0.9 under truthful communication and around 0.8 in the data. When both other members acquire, cursed message inference leaves the value of acquisition unchanged from the benchmark. Without one's own acquisition, group accuracy is q; with it, the collective decision follows the majority of the three signals, yielding accuracy $3q^2 - 2q^3$. Hence $\widetilde{\Delta}_2^C(\chi) = (2q-1)q(1-q) = \Delta_2^{NC}$ for every $\chi \in [0,1]$.

[34] More generally, with independent, equally precise signals and random tie-breaking, adding one signal to an odd number of signals does not change expected decision accuracy. This odd–even property also appears in the Condorcet jury literature; see, for example, Berend and Sapir (2005). In my setting, with $N = 3$, the substitution effect arises only in the contingency in which one other member acquires, which operates entirely through the odd–even property. In larger groups, the substitution effect also arises in contingencies that do not rely on this property. The $N = 3$ parameterization is therefore particular in this respect.

this possibility. Aycinena et al. (2023) study a binary-state information-acquisition task in which purchasing a positive even number of signals is dominated because the final signal cannot make the decision maker strictly prefer a different action; nevertheless, subjects frequently purchase even numbers of signals. More generally, Rieger et al. (2021) find substantial over-acquisition of information and attribute it primarily to subjects overestimating the probability that additional information will improve the eventual decision.

The two interpretations locate the missing substitution response at different points in the reasoning process. Under-inference from communication makes the other member's communicated information appear less informative than it is; under overvaluation, the other member's information may be correctly understood, but the incremental decision value of one's own signal is overestimated. In either case, subjects insufficiently account for the communication-induced reduction in the value of their own information.[35] Both plausibly arise outside the laboratory: members of actual committees likewise need to draw inferences from others' statements and to judge the value of additional information.

6 Conclusion

This paper asks how pre-vote communication affects costly information acquisition in committees. In a common-interest majority-voting model, the theory identifies two opposing forces created by a pre-vote communication stage. Communication raises the return to acquiring one's own signal by letting it influence the group—the persuasion effect—but also weakens that incentive by making information acquired by other members more influential—the substitution effect. In equilibrium, communication mitigates free riding when information is relatively expensive but can exacerbate it when information is relatively cheap. Group accuracy and welfare exhibit an analogous cost-dependent pattern. A laboratory experiment with three-member groups and two information costs reveals only the beneficial side of this theoretical tradeoff: acquisition is higher under communication at both cost levels, and communication improves group accuracy and welfare even at the low cost, where the theory predicts declines in all three outcomes.

Subjects' stated conditional choices help explain why the predicted amplification of free riding does not emerge: they respond strongly when communication makes their own signal more valuable, but hardly at all when it makes that signal less valuable. That is,

[35]Separating the two interpretations would require a different design. One possibility would be to remove the need to infer the informativeness of others' messages, for example by making acquired signals directly observable. Alternatively, varying signal precision could reveal whether subjects systematically overvalue the marginal decision value of additional information; perfectly informative signals provide a particularly transparent case.

they appear to recognize the persuasion effect but insufficiently account for the substitution effect. Consistent with this asymmetry, at both cost levels subjects overestimate the value of their own information acquisition under communication relative to its empirical value. They also perceive acquisition as more valuable with communication than without, even at the low cost where communication in fact lowers that value. This weak response to the substitution effect admits two interpretations: subjects may treat others' messages as less informative than they are, or they may overvalue the incremental contribution of an additional noisy signal.

These findings have two implications for the design and evaluation of committee decision-making procedures. First, alongside voting rules and committee size, whether to allow pre-vote communication is itself an institutional margin for shaping incentives to acquire information. This margin may be particularly relevant when voting rules and committee size are fixed by law, convention, or organizational structure, while the difficulty of becoming informed varies across decisions. In such settings, deliberation provides an additional, potentially more flexible way to shape information acquisition.

Second, the findings add an information-acquisition dimension to the debate over whether and when deliberation improves collective decisions.[36] The theory points to a caveat: allowing members to talk can be harmful, as it may amplify free riding in information provision. The experiment, however, suggests that this downside may be behaviorally attenuated when individuals do not fully internalize the additional substitution incentive created by communication. Whether the theoretical downside emerges in other environments, such as larger committees or settings with verifiable disclosure of acquired information, remains an open question. Evaluating the effects of deliberation therefore requires attention not only to how communication aggregates existing information, but also to how it changes—and how individuals perceive it to change—the incentives to acquire information in the first place.

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[36]For example, Iaryczower et al. (2018) structurally estimate the effect of deliberation in U.S. appellate courts by comparing decisions under deliberation with a counterfactual without communication. Their model holds judges' private information structure fixed across the two regimes. The results here suggest that the amount of private information held by members may itself vary with deliberation, potentially affecting such counterfactual evaluations.

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Appendix

A Closed-form Expressions

This appendix collects the closed-form expressions and auxiliary formulas used in the theoretical analysis. I first derive the value of information acquisition in games NC and C, then provide explicit expressions for the persuasion and substitution terms, and finally present the group-accuracy formulas used in the analysis.

A.1 Value of information acquisition in game NC

Fix voter i, and suppose that every other voter $j \neq i$ acquires information independently with probability $\alpha \in [0,1]$. Under sincere voting in game NC, voter j votes for the correct alternative with probability

$$p(\alpha) := \alpha q + \tfrac{1}{2}(1-\alpha) = \tfrac{1}{2} + \alpha\big(q - \tfrac{1}{2}\big).$$

Conditional on $z = A$, each of the $2n$ voters other than i therefore votes for A with probability $p(\alpha)$ and for B with probability $1 - p(\alpha)$, independently. Hence

$$V_A^{-i} \sim \mathrm{Bin}(2n, p(\alpha)).$$

By symmetry, the same pivotal probability obtains conditional on $z = B$. Since $V_A^{-i} = V_B^{-i}$ if and only if $V_A^{-i} = n$,

$$\Pr(V_A^{-i} = V_B^{-i}) = \binom{2n}{n} p(\alpha)^n \big(1 - p(\alpha)\big)^n.$$

Substituting this expression into equation (1) gives

$$\Delta^{NC}(\alpha) = \left(q - \frac{1}{2}\right) \binom{2n}{n} \left(q\alpha + \frac{1}{2}(1-\alpha)\right)^n \left(1 - q\alpha - \frac{1}{2}(1-\alpha)\right)^n. \tag{4}$$

A.2 Value of information acquisition in game C

Again fix voter i, and suppose that every other voter $j \neq i$ acquires information independently with probability $\alpha \in [0,1]$. Let K denote the number of acquirers among the other $2n$ voters. Then $K \sim \mathrm{Bin}(2n, \alpha)$. The information-pivotal event $S_a^{-i} = S_b^{-i}$ can occur only

when K is even. In particular,

$$\Pr(K = 2k) = \binom{2n}{2k}\alpha^{2k}(1-\alpha)^{2n-2k}, \qquad k = 0, \ldots, n.$$

Conditional on $z = A$ and $K = 2k$, the number of a signals is distributed as $\text{Bin}(2k, q)$. Hence,

$$\Pr(S_a^{-i} = S_b^{-i} \mid K = 2k) = \binom{2k}{k}q^k(1-q)^k.$$

By the law of total probability,

$$\Pr(S_a^{-i} = S_b^{-i}) = \sum_{k=0}^{n}\binom{2n}{2k}\alpha^{2k}(1-\alpha)^{2n-2k}\binom{2k}{k}q^k(1-q)^k.$$

Substituting into equation (2) gives

$$\Delta^C(\alpha) = \left(q - \frac{1}{2}\right)\sum_{k=0}^{n}\binom{2n}{2k}\alpha^{2k}(1-\alpha)^{2n-2k}\binom{2k}{k}q^k(1-q)^k. \tag{5}$$

A.3 Persuasion and substitution effects

The decomposition in Section 2.2 writes the effect of communication on the value of information acquisition as

$$\Delta^C(\alpha) - \Delta^{NC}(\alpha) = \left(q - \tfrac{1}{2}\right)\big(P(\alpha) - S(\alpha)\big),$$

where

$$P(\alpha) := \Pr\Big((S_a^{-i} = S_b^{-i}) \cap (U_A^{-i} \neq U_B^{-i})\Big)$$

is the persuasion term and

$$S(\alpha) := \Pr\Big((S_a^{-i} \neq S_b^{-i}) \cap (S_a^{-i} + U_A^{-i} = S_b^{-i} + U_B^{-i})\Big)$$

is the substitution term. These probabilities admit the following finite-sum expressions.

For the persuasion term, conditional on $K = 2k$, the acquired signals must split evenly, while the $2n - 2k$ uninformed votes must fail to split evenly. Therefore,

$$\begin{aligned} P(\alpha) = \sum_{k=0}^{n}\binom{2n}{2k}\alpha^{2k}(1-\alpha)^{2n-2k}\binom{2k}{k}\big(q(1-q)\big)^k \\ \times\left[1 - \binom{2n-2k}{n-k}2^{-(2n-2k)}\right]. \end{aligned} \tag{6}$$

For the substitution term, let K be the number of acquirers among the other $2n$ voters and let s be the number of a signals among them. Conditional on $z = A$, the signal profile has probability

$$\binom{K}{s} q^s (1-q)^{K-s}.$$

For voter i to be vote-pivotal in game NC, the $2n-K$ uninformed voters must cast exactly $n-s$ votes for A. Excluding the information-pivotal case $s = K/2$ gives

$$\begin{aligned} S(\alpha) = \sum_{K=0}^{2n} \binom{2n}{K} \alpha^K (1-\alpha)^{2n-K} \sum_{s=\max\{0,K-n\}}^{\min\{K,n\}} \mathbf{1}\{s \neq K/2\} \\ \times \binom{K}{s} q^s (1-q)^{K-s} \binom{2n-K}{n-s} 2^{-(2n-K)}. \end{aligned} \tag{7}$$

A.4 Group accuracy

For a fixed symmetric acquisition probability α, define

$$p(\alpha) = \alpha q + \frac{1}{2}(1-\alpha),$$

the probability that an individual vote is correct in game NC. Since the group contains $2n+1$ voters, the collective decision is correct if and only if at least $n+1$ votes are correct. Hence,

$$G^{NC}(\alpha) = \sum_{k=n+1}^{2n+1} \binom{2n+1}{k} p(\alpha)^k \big(1-p(\alpha)\big)^{2n+1-k}. \tag{8}$$

In game C, conditional on either state, each voter independently contributes a correct signal with probability αq, an incorrect signal with probability $\alpha(1-q)$, and no signal with probability $1-\alpha$. Under full revelation, the collective decision follows the majority of the acquired signals, with a tie yielding accuracy $1/2$. Therefore,

$$\begin{aligned} G^{C}(\alpha) = \sum_{k=1}^{2n+1} \binom{2n+1}{k} (\alpha q)^k \sum_{y=0}^{\min\{k-1,\,2n+1-k\}} \binom{2n+1-k}{y} \big(\alpha(1-q)\big)^y (1-\alpha)^{2n+1-k-y} \\ + \frac{1}{2} \sum_{k=0}^{n} \binom{2n+1}{k} \binom{2n+1-k}{k} (\alpha q)^k \big(\alpha(1-q)\big)^k (1-\alpha)^{2n+1-2k}. \end{aligned} \tag{9}$$

The first term collects realizations in which correct signals strictly outnumber incorrect signals, while the second collects ties between the two signal realizations.

B Proofs

B.1 Proof for Post-acquisition Behavior

Proof of Lemma 1. We prove the two parts in turn.

Game NC. Fix voter i. By symmetry, let

$$p := \Pr(v_j = A \mid z = A)$$

for any voter $j \neq i$. Neutrality implies

$$\Pr(v_j = A \mid z = B) = 1 - p.$$

Hence, the probability that the other $2n$ voters are evenly split between A and B is

$$\binom{2n}{n} p^n (1-p)^n$$

under either state. Thus, conditional on being pivotal, voter i learns nothing additional about the state beyond his or her own type.

It follows that, when pivotal, a voter with signal a prefers A, a voter with signal b prefers B, and an uninformed voter is indifferent between the two alternatives. When not pivotal, the vote does not affect the collective decision. Therefore voting is sincere. By neutrality, an uninformed voter randomizes uniformly between A and B.

Game C. We proceed in two steps.

Step 1: A fully revealing PBE with sincere voting exists. Choose three distinct messages in M, one for each type in T. Consider a strategy profile in which each voter truthfully reveals his or her type and, following truthful revelation, votes sincerely, randomizing uniformly when the common posterior is $1/2$.

First consider the voting stage. Under truthful revelation, the realized type profile is common knowledge, so all voters share the same posterior. Voting for the posterior-favored alternative is therefore optimal; when the posterior is $1/2$, either vote is optimal.

Next consider the communication stage. Fix any realized type profile t. Under truthful revelation, the collective decision maximizes the probability of matching the state conditional on t. Any unilateral message deviation can induce only some other decision, or a lottery over the two alternatives, and therefore cannot yield a higher probability of matching the state conditional on t. Truthful revelation is thus optimal for every type.

Finally, given these continuation strategies, the information-acquisition stage is a finite symmetric game and therefore admits a symmetric mixed-strategy equilibrium. Take any such equilibrium. Together with the communication and voting strategies above, and any symmetric and neutral off-path beliefs and continuation strategies that satisfy sequential rationality, this gives a symmetric and neutral PBE with full revelation and sincere voting.

Step 2: The equilibrium restrictions imply full revelation and sincere voting. Step 1 shows that full revelation of the realized post-acquisition type profile is attainable in equilibrium. Since no communication outcome can reveal more than the entire type profile, maximal information transmission implies full revelation in the equilibria considered in the paper.
Under full revelation, the realized type profile is common knowledge, so all voters share the same posterior. If the posterior strictly favors A, then voting for A weakly dominates voting for B: the two votes yield the same outcome whenever the voter is not pivotal, while voting for A yields a strictly higher expected payoff whenever he or she is pivotal. The analogous argument applies when the posterior favors B. When the posterior is $1/2$, either vote is sincere. Hence, the restriction to weakly undominated voting strategies implies sincere voting.
This proves both parts. □

B.2 Proof for Information Acquisition

Throughout this subsection, fix $n \geq 1$ and $q \in (\frac{1}{2}, 1)$, and let

$$\delta := q - \tfrac{1}{2} \in \left(0, \tfrac{1}{2}\right), \qquad r := q(1-q) = \tfrac{1}{4} - \delta^2 \in \left(0, \tfrac{1}{4}\right).$$

Proof of Lemma 2. *1. Reparameterization.* For $\alpha \in [0, 1)$, set

$$x := \frac{\alpha}{1-\alpha} \in [0, \infty), \qquad \alpha = \frac{x}{1+x}, \qquad 1 - \alpha = \frac{1}{1+x}.$$

The map $\alpha \mapsto x$ is a strictly increasing bijection from $[0, 1)$ onto $[0, \infty)$. Equations (5) and (4) become

$$\Delta^C(\alpha) = \delta(1+x)^{-2n} \sum_{k=0}^{n} \binom{2n}{2k} \binom{2k}{k} (rx^2)^k$$

and

$$\Delta^{NC}(\alpha) = \delta(1+x)^{-2n} \binom{2n}{n} \left(\tfrac{1}{4} + \tfrac{x}{2} + rx^2\right)^n.$$

Define

$$S(u) := \sum_{k=0}^{n} \binom{2n}{2k}\binom{2k}{k} u^k, \qquad F(x) := S(rx^2) - \binom{2n}{n}\left(\tfrac{1}{4} + \tfrac{x}{2} + rx^2\right)^n.$$

Then

$$\Delta^C(\alpha) - \Delta^{NC}(\alpha) = \delta(1+x)^{-2n} F(x),$$

so the sign of $\Delta^C(\alpha) - \Delta^{NC}(\alpha)$ equals the sign of $F(x)$.
At $x = 0$,

$$F(0) = 1 - \frac{\binom{2n}{n}}{4^n} > 0,$$

because $\binom{2n}{n} < \sum_{j=0}^{2n}\binom{2n}{j} = 4^n$. As $x \to \infty$,

$$S(rx^2) = \binom{2n}{n} r^n x^{2n} + O(x^{2n-2}), \qquad \left(\tfrac{1}{4} + \tfrac{x}{2} + rx^2\right)^n = r^n x^{2n} + \frac{n}{2} r^{n-1} x^{2n-1} + O(x^{2n-2}),$$

so

$$F(x) = -\binom{2n}{n}\frac{n}{2} r^{n-1} x^{2n-1} + O(x^{2n-2}) \to -\infty.$$

Hence there exists $X > 0$ such that $F(x) < 0$ for all $x \geq X$.

2. Integral representation. Using $\int_0^\pi \cos^j\theta\, d\theta = 0$ for odd j and

$$\int_0^\pi \cos^{2k}\theta\, d\theta = \pi \frac{\binom{2k}{k}}{4^k},$$

a binomial expansion gives

$$S(u) = \frac{1}{\pi}\int_0^\pi (1 + 2\sqrt{u}\cos\theta)^{2n}\, d\theta, \qquad u \geq 0. \tag{10}$$

Define the normalized L^{2n}-norm on $[0,\pi]$ by

$$\|f\|_{2n} := \left(\frac{1}{\pi}\int_0^\pi |f(\theta)|^{2n}\, d\theta\right)^{1/(2n)}.$$

Let

$$t := \sqrt{r}\, x, \qquad Q(t) := t^2 + \frac{t}{2\sqrt{r}} + \frac{1}{4}, \qquad A_n := \binom{2n}{n}^{1/(2n)},$$

and define, for $t \geq 0$,

$$\Phi(t) := \|1 + 2t\cos\theta\|_{2n}, \qquad \Psi(t) := A_n\sqrt{Q(t)}.$$

Since $rx^2 = t^2$ and $x/2 = t/(2\sqrt{r})$, equation (10) yields

$$F(x) = \Phi(t)^{2n} - \Psi(t)^{2n}.$$

Because $\Phi, \Psi > 0$,

$$\operatorname{sign} F(x) = \operatorname{sign}\big(\Phi(t) - \Psi(t)\big).$$

3. Curvature of Φ and Ψ. Let $t_1 \neq t_2$ and $\lambda \in (0,1)$. By Minkowski's inequality,

$$\Phi\big(\lambda t_1 + (1-\lambda)t_2\big) \leq \lambda\Phi(t_1) + (1-\lambda)\Phi(t_2).$$

Since $2n > 1$, equality would require $1 + 2t_1\cos\theta = \mu(1 + 2t_2\cos\theta)$ for almost every θ and some $\mu \geq 0$. Integrating both sides over $[0,\pi]$ gives $\mu = 1$, and then $t_1 = t_2$, a contradiction. Hence Φ is strictly convex.
Write $Q(t) = t^2 + at + b$ with $a := 1/(2\sqrt{r})$ and $b := 1/4$. Then

$$\frac{d^2}{dt^2}\sqrt{Q(t)} = \frac{4b - a^2}{4Q(t)^{3/2}} = \frac{1 - \frac{1}{4r}}{4Q(t)^{3/2}} < 0,$$

since $r < \frac{1}{4}$. Hence Ψ is strictly concave.

4. Unique crossing. Let $h(t) := \Phi(t) - \Psi(t)$ for $t \geq 0$. By Step 3, h is continuous and strictly convex. Moreover,

$$h(0) = 1 - \frac{A_n}{2} > 0,$$

because $A_n < 2$, and $h(t) < 0$ for all $t \geq T := \sqrt{r}\,X$ by Step 1.
By the intermediate value theorem, h has a zero in $(0,T)$. This zero is unique. Suppose instead that $h(t_1) = h(t_2) = 0$ for some $0 < t_1 < t_2$. Pick $t_3 > \max\{t_2, T\}$ and write $t_2 = \lambda t_1 + (1-\lambda)t_3$ with $\lambda \in (0,1)$. Strict convexity gives

$$0 = h(t_2) < \lambda h(t_1) + (1-\lambda)h(t_3) = (1-\lambda)h(t_3) < 0,$$

a contradiction. Denote the unique zero by $t^* > 0$. Since h is continuous with no other zero, $h(0) > 0$, and $h < 0$ on $[T,\infty)$, we obtain

$$h(t) > 0 \ \text{ for } t \in [0, t^*), \qquad h(t) < 0 \ \text{ for } t \in (t^*, \infty).$$

5. Return to α. Let $x^* := t^*/\sqrt{r}$ and

$$\hat{\alpha} := \frac{x^*}{1+x^*} \in (0,1).$$

Since $\alpha \mapsto t = \sqrt{r}\,\alpha/(1-\alpha)$ is a strictly increasing bijection from $[0,1)$ onto $[0,\infty)$, Steps 1, 2 and 4 give

$$\Delta^C(\alpha) > \Delta^{NC}(\alpha) \text{ for } \alpha \in [0,\hat{\alpha}), \qquad \Delta^C(\hat{\alpha}) = \Delta^{NC}(\hat{\alpha}), \qquad \Delta^C(\alpha) < \Delta^{NC}(\alpha) \text{ for } \alpha \in (\hat{\alpha},1).$$

Finally, letting $\alpha \to 1$ (equivalently, $x \to \infty$) in the expressions of Step 1 and using continuity of Δ^C and Δ^{NC} in α,

$$\Delta^C(1) = \Delta^{NC}(1) = \delta\binom{2n}{n} r^n.$$

This proves the result. □

Lemma 4. *$\Delta^{NC}(\alpha)$ is strictly decreasing in $\alpha \in [0,1]$. $\Delta^C(\alpha)$ is strictly convex in $\alpha \in [0,1]$ and has a unique minimizer $\underline{\alpha} \in (0,1)$; it is strictly decreasing on $[0,\underline{\alpha}]$ and strictly increasing on $[\underline{\alpha},1]$.*

Proof. For game NC, substituting $x = \alpha/(1-\alpha)$ into the expression for Δ^{NC} in Step 1 of the proof of Lemma 2 gives

$$\Delta^{NC}(\alpha) = \delta\binom{2n}{n}\left(\frac{1}{4} - \delta^2\alpha^2\right)^n,$$

so

$$\frac{d}{d\alpha}\Delta^{NC}(\alpha) = -2n\delta^3\alpha\binom{2n}{n}\left(\frac{1}{4} - \delta^2\alpha^2\right)^{n-1} < 0 \qquad \text{for all } \alpha \in (0,1].$$

Hence Δ^{NC} is strictly decreasing on $[0,1]$.
For game C, equation (10) gives

$$\Delta^C(\alpha) = \frac{\delta}{\pi}\int_0^\pi g_\theta(\alpha)^{2n}\,d\theta, \qquad g_\theta(\alpha) := 1 - \alpha + 2\alpha\sqrt{r}\cos\theta.$$

Since $g'_\theta(\alpha) = -1 + 2\sqrt{r}\cos\theta$,

$$\frac{d^2}{d\alpha^2}\Delta^C(\alpha) = \frac{2n(2n-1)\delta}{\pi}\int_0^\pi g_\theta(\alpha)^{2n-2}\left(-1+2\sqrt{r}\cos\theta\right)^2 d\theta > 0,$$

because $2\sqrt{r} < 1$ implies $-1 + 2\sqrt{r}\cos\theta < 0$ for every θ, and $g_\theta(\alpha)^{2n-2} > 0$ for almost every

θ. Hence Δ^C is strictly convex on $[0,1]$. Moreover,

$$\frac{d}{d\alpha}\Delta^C(0) = \frac{2n\delta}{\pi}\int_0^\pi \left(-1 + 2\sqrt{r}\cos\theta\right) d\theta = -2n\delta < 0,$$

whereas, since $\int_0^\pi \cos^{2n-1}\theta\, d\theta = 0$,

$$\frac{d}{d\alpha}\Delta^C(1) = \frac{2n\delta}{\pi}(2\sqrt{r})^{2n}\int_0^\pi \cos^{2n}\theta\, d\theta > 0.$$

Since the derivative of Δ^C is continuous and strictly increasing, it has a unique zero $\underline{\alpha} \in (0,1)$, is negative on $[0,\underline{\alpha})$, and is positive on $(\underline{\alpha},1]$. This proves the claim. □

Proof of Lemma 3. Fix $X \in \{NC, C\}$. Since voter i's best response is to acquire if $\Delta^X(\alpha) > c$, not to acquire if $\Delta^X(\alpha) < c$, and to mix arbitrarily if $\Delta^X(\alpha) = c$, a symmetric profile α is an equilibrium of game X if and only if

(i) $\alpha = 0$ and $\Delta^X(0) \le c$; or

(ii) $\alpha = 1$ and $\Delta^X(1) \ge c$; or

(iii) $\alpha \in (0,1)$ and $\Delta^X(\alpha) = c$.

Preparations.

(P1) $\Delta^C(0) > \Delta^{NC}(0)$, by Lemma 2.

(P2) By Lemma 4, $\Delta^{NC}(0) > \Delta^{NC}(\alpha) > \Delta(1)$ for all $\alpha \in (0,1)$.

(P3) By (P1) and (P2), $\Delta^C(0) > \Delta^{NC}(0) > \Delta(1) = \Delta^C(1) > \Delta^C(\underline{\alpha})$, where the last inequality holds because $\underline{\alpha}$ is the unique minimizer of Δ^C. By Lemma 4, Δ^C falls strictly from $\Delta^C(0)$ to $\Delta^C(\underline{\alpha})$ on $[0,\underline{\alpha}]$ and rises strictly from $\Delta^C(\underline{\alpha})$ to $\Delta(1)$ on $[\underline{\alpha},1]$. Hence there is a unique $\alpha' \in (0,\underline{\alpha})$ with $\Delta^C(\alpha') = \Delta(1)$, and

$$\Delta(1) < \Delta^C(\alpha) < \Delta^C(0) \ \text{ for } \alpha \in (0,\alpha'), \qquad \Delta^C(\alpha) < \Delta(1) \ \text{ for } \alpha \in (\alpha',1).$$

In particular, $\Delta^C(\alpha) < \Delta^C(0)$ for all $\alpha \in (0,1]$.

(P4) Since $\hat{\alpha} \in (0,1)$, (P2) gives $\Delta(1) < \hat{c} < \Delta^{NC}(0)$. Since $\Delta^C(\hat{\alpha}) = \hat{c} > \Delta(1)$, (P3) gives $\hat{\alpha} < \alpha'$.

1. Suppose $c \ge \Delta^C(0)$. By (P1)–(P3), $c > \Delta^X(\alpha)$ for all $\alpha \in (0,1]$ and $X \in \{NC, C\}$, so conditions (ii) and (iii) fail, while (i) holds. Hence $\alpha^{*NC} = \alpha^{*C} = 0$ is the unique equilibrium of each game.

2. Suppose $c \in (\Delta(1), \Delta^C(0))$.

Game C. Conditions (i) and (ii) fail because $\Delta^C(0) > c > \Delta(1) = \Delta^C(1)$. By (P3), $\Delta^C(\alpha) \leq \Delta(1) < c$ for $\alpha \in [\alpha', 1)$, so any solution to $\Delta^C(\alpha) = c$ lies in $(0, \alpha')$. On $[0, \alpha']$, Δ^C is continuous and strictly decreasing from $\Delta^C(0)$ to $\Delta(1)$, so there is exactly one such solution. Hence game C has a unique equilibrium, and $\alpha^{*C} \in (0, \alpha')$.

Game NC. Condition (ii) fails because $\Delta^{NC}(1) = \Delta(1) < c$. If $c \geq \Delta^{NC}(0)$, (P2) implies that (iii) fails while (i) holds, so $\alpha^{*NC} = 0$ is the unique equilibrium. If $c < \Delta^{NC}(0)$, (i) fails, and since Δ^{NC} is continuous and strictly decreasing from $\Delta^{NC}(0)$ to $\Delta(1)$, there is exactly one $\alpha \in (0, 1)$ with $\Delta^{NC}(\alpha) = c$. Hence game NC has a unique equilibrium, and $\alpha^{*NC} \in [0, 1)$.

Comparison. Recall from (P4) that $\Delta(1) < \hat{c} < \Delta^{NC}(0) < \Delta^C(0)$.

(a) If $c \in [\Delta^{NC}(0), \Delta^C(0))$, then $\alpha^{*NC} = 0 < \alpha^{*C}$.

(b) If $c \in (\hat{c}, \Delta^{NC}(0))$, then $\Delta^{NC}(\alpha^{*NC}) = c > \hat{c} = \Delta^{NC}(\hat{\alpha})$, so $\alpha^{*NC} < \hat{\alpha}$ because Δ^{NC} is strictly decreasing. By Lemma 2,

$$\Delta^C(\alpha^{*NC}) > \Delta^{NC}(\alpha^{*NC}) = c = \Delta^C(\alpha^{*C}).$$

Since $\alpha^{*NC}, \alpha^{*C} \in [0, \alpha')$ and Δ^C is strictly decreasing on $[0, \alpha']$, it follows that $\alpha^{*C} > \alpha^{*NC}$.

(c) If $c = \hat{c}$, then $\Delta^{NC}(\hat{\alpha}) = \Delta^C(\hat{\alpha}) = c$ with $\hat{\alpha} \in (0, \alpha')$ by (P4), so uniqueness in each game gives $\alpha^{*C} = \alpha^{*NC} = \hat{\alpha}$.

(d) If $c \in (\Delta(1), \hat{c})$, then $\Delta^{NC}(\alpha^{*NC}) = c < \hat{c}$, so $\alpha^{*NC} \in (\hat{\alpha}, 1)$. By Lemma 2,

$$\Delta^C(\alpha^{*NC}) < \Delta^{NC}(\alpha^{*NC}) = c = \Delta^C(\alpha^{*C}).$$

If $\alpha^{*NC} \geq \alpha'$, then $\alpha^{*C} < \alpha' \leq \alpha^{*NC}$. If $\alpha^{*NC} < \alpha'$, then $\alpha^{*NC}, \alpha^{*C} \in (0, \alpha')$, and strict decrease of Δ^C on $[0, \alpha']$ gives $\alpha^{*C} < \alpha^{*NC}$.

Thus $\alpha^{*C} > \alpha^{*NC}$ if $c > \hat{c}$, $\alpha^{*C} < \alpha^{*NC}$ if $c < \hat{c}$, and $\alpha^{*C} = \alpha^{*NC} = \hat{\alpha}$ if $c = \hat{c}$.

3. Suppose $c \leq \Delta(1)$.

Game NC. Condition (ii) holds because $\Delta^{NC}(1) = \Delta(1) \geq c$. By (P2), $\Delta^{NC}(\alpha) > \Delta(1) \geq c$ for all $\alpha \in [0, 1)$, so (i) and (iii) fail. Hence $\alpha^{*NC} = 1$ is the unique equilibrium.

Game C. Condition (ii) holds, so $\alpha^{*C} = 1$ is an equilibrium. Condition (i) fails because $\Delta^C(0) > \Delta(1) \geq c$. By (P3) and Lemma 4, the interior solutions to $\Delta^C(\alpha) = c$ are as follows:

- if $c = \Delta(1)$, the unique interior solution is α';
- if $c \in (\Delta^C(\underline{\alpha}), \Delta(1))$, there are exactly two interior solutions, one in $(\alpha', \underline{\alpha})$ and one in $(\underline{\alpha}, 1)$;
- if $c = \Delta^C(\underline{\alpha})$, the unique interior solution is $\underline{\alpha}$;
- if $c < \Delta^C(\underline{\alpha})$, there is no interior solution.

Thus game C always admits $\alpha^{*C} = 1$ and, for $c \in [\Delta^C(\underline{\alpha}), \Delta(1)]$, also admits equilibria with $\alpha^{*C} < 1$. □

Proof of Proposition 1. By (P4) in the proof of Lemma 3, $\hat{c} \in (\Delta(1), \Delta^{NC}(0)) \subset (\Delta(1), \Delta^C(0))$.

$\hat{c}$ is a cutoff. If $c \geq \Delta^C(0)$, Lemma 3(1) gives $\alpha^{*C} = \alpha^{*NC} = 0$, so both weak inequalities hold. If $c \in (\Delta(1), \Delta^C(0))$, the proof of Lemma 3(2) gives $\alpha^{*C} > \alpha^{*NC}$ for $c > \hat{c}$, $\alpha^{*C} < \alpha^{*NC}$ for $c < \hat{c}$, and $\alpha^{*C} = \alpha^{*NC} = \hat{\alpha}$ for $c = \hat{c}$. If $c \leq \Delta(1)$, Lemma 3(3) gives $\alpha^{*NC} = 1 \geq \alpha^{*C}$ for every equilibrium $\alpha^{*C} \in [0, 1]$ of game C. Hence $\alpha^{*C} \geq \alpha^{*NC}$ for $c \geq \hat{c}$ and $\alpha^{*C} \leq \alpha^{*NC}$ for $c \leq \hat{c}$, with strict inequality for $c \in (\Delta(1), \Delta^C(0))$ with $c \neq \hat{c}$.

Uniqueness. Suppose some $c' \neq \hat{c}$ also has the cutoff property. If $c' > \hat{c}$, take $c \in (\hat{c}, \min\{c', \Delta^C(0)\})$. Since $c \leq c'$, the property requires $\alpha^{*C} \leq \alpha^{*NC}$, contradicting $\alpha^{*C} > \alpha^{*NC}$. If $c' < \hat{c}$, take $c \in (\max\{c', \Delta(1)\}, \hat{c})$. Since $c \geq c'$, the property requires $\alpha^{*C} \geq \alpha^{*NC}$, contradicting $\alpha^{*C} < \alpha^{*NC}$. Hence $\hat{c}$ is the unique cutoff. □

B.3 Proof for Group Accuracy

We first establish two preliminary facts used in the proof of Proposition 2.

Fixed-acquisition comparison. We first verify equation (3). By symmetry, it suffices to condition on one state. Let K, Y, and U denote the numbers of informed voters receiving the correct signal, informed voters receiving the incorrect signal, and uninformed voters, respectively. For any $k + y + u = N$,

$$\Pr(K = k, Y = y, U = u) = \frac{N!}{k!\, y!\, u!}(\alpha q)^k \big(\alpha(1-q)\big)^y (1-\alpha)^u.$$

Let $c(k, y, u)$ and $s(k, y, u)$ denote the conditional probabilities of a correct collective decision in games C and NC. Under communication, $c(k, y, u) = 1$ if $k > y$, $c(k, y, u) = 1/2$ if $k = y$, and $c(k, y, u) = 0$ if $k < y$. Hence $c(k, y, u) + c(y, k, u) = 1$. Likewise, exchanging correct and incorrect informed signals in game NC complements the event that the collective decision is correct, so $s(k, y, u) + s(y, k, u) = 1$.

Pairing the terms (k, y, u) and (y, k, u) for $k > y$ in $G^C(\alpha) - G^{NC}(\alpha)$, while terms with $k = y$ cancel, gives the combined contribution

$$\big(\Pr(k, y, u) - \Pr(y, k, u)\big)\big(1 - s(k, y, u)\big).$$

This is nonnegative because $1 - s(k, y, u) \geq 0$ and

$$\frac{\Pr(k, y, u)}{\Pr(y, k, u)} = \left(\frac{q}{1-q}\right)^{k-y} > 1.$$

Therefore $G^C(\alpha) \geq G^{NC}(\alpha)$. Equality holds at $\alpha = 0$ and $\alpha = 1$. For $\alpha \in (0, 1)$, the profile $(k, y, u) = (1, 0, N-1)$ occurs with positive probability and has $s(k, y, u) < 1$, so the inequality is strict.

Accuracy and the value of acquisition. Let $v^X(K)$ denote group accuracy when exactly K voters acquire information. Since $G^X(\alpha) = \mathbb{E}[v^X(K)]$ with $K \sim \text{Bin}(N, \alpha)$, the derivative identity for binomial expectations and voter symmetry give

$$G^{X\prime}(\alpha) = N\Delta^X(\alpha). \tag{11}$$

Since $G^X(0) = 1/2$,

$$G^X(\alpha) = \frac{1}{2} + N\int_0^\alpha \Delta^X(t)\, dt. \tag{12}$$

In particular, G^X is strictly increasing.

We next establish one auxiliary property used below and again in the proof of Proposition 3.

Lemma 5. *For $\alpha \in (\hat{\alpha}, 1)$, let $\beta(\alpha)$ be defined by $\Delta^C(\beta(\alpha)) = \Delta^{NC}(\alpha)$ on the decreasing branch of Δ^C. Then $\beta(\alpha)$ is strictly convex.*

Proof. Let $d := q - \frac{1}{2}$, and write $f := \Delta^C/d$ and $g := \Delta^{NC}/d$. Since $f(\beta) = g(\alpha)$,

$$\beta'(\alpha) = \frac{g'(\alpha)}{f'(\beta)}, \qquad \beta''(\alpha) = \frac{g''(\alpha)f'(\beta)^2 - g'(\alpha)^2 f''(\beta)}{f'(\beta)^3}.$$

Because β lies on the decreasing branch of f, $f'(\beta) < 0$. Hence $\beta''(\alpha) > 0$ if and only if

$$\frac{g''(\alpha)}{g'(\alpha)^2} < \frac{f''(\beta)}{f'(\beta)^2}. \tag{13}$$

If $g''(\alpha) \leq 0$, this follows immediately from $f''(\beta) > 0$. Suppose therefore that $g''(\alpha) > 0$.

Using the integral representation established above, write

$$f(\beta) = \frac{1}{\pi}\int_0^\pi h^{2n}\,d\theta, \qquad h := 1-\beta+2\beta\sqrt{r}\cos\theta, \qquad w := 2\sqrt{r}\cos\theta - 1.$$

Then

$$f'(\beta) = \frac{2n}{\pi}\int_0^\pi h^{2n-1} w\,d\theta, \qquad f''(\beta) = \frac{2n(2n-1)}{\pi}\int_0^\pi h^{2n-2}w^2\,d\theta.$$

Cauchy–Schwarz therefore implies

$$\frac{f''(\beta)}{f'(\beta)^2} \geq \frac{2n-1}{2n\,f(\beta)} = \frac{2n-1}{2n\,g(\alpha)}. \tag{14}$$

Thus (13) follows if

$$(2n-1)g'(\alpha)^2 > 2n\,g(\alpha)g''(\alpha). \tag{15}$$

Let $C := \binom{2n}{n}$ and $v := \frac{1}{4} - d^2\alpha^2$, so $g(\alpha) = Cv^n$. Then

$$g'(\alpha) = -2Cnd^2\alpha v^{n-1}, \qquad g''(\alpha) = -2Cnd^2 v^{n-2}\big[v - 2(n-1)d^2\alpha^2\big].$$

Hence

$$(2n-1)g'(\alpha)^2 - 2n\,g(\alpha)g''(\alpha) = 4C^2n^2d^2v^{2n-2}\big[d^2\alpha^2 + v\big] = C^2n^2d^2v^{2n-2} > 0,$$

where the second equality uses $d^2\alpha^2 + v = 1/4$. Thus (15) holds and $\beta''(\alpha) > 0$. □

Proof of Proposition 2. Suppose first that $c \geq \hat{c}$. By Proposition 1, $\alpha^{*C} \geq \alpha^{*NC}$. Since G^C is increasing and (3) holds,

$$G^{*C} = G^C(\alpha^{*C}) \geq G^C(\alpha^{*NC}) \geq G^{NC}(\alpha^{*NC}) = G^{*NC}.$$

The inequality is strict for $c \in [\hat{c}, \Delta^C(0))$: if the equilibrium acquisition rates differ, the first inequality is strict; at $c = \hat{c}$, they coincide at $\hat{\alpha} \in (0,1)$ and the second inequality is strict by (3).

If $c \leq \Delta(1)$, game NC has the unique equilibrium $\alpha^{*NC} = 1$, while game C always admits $\alpha^{*C} = 1$ and may admit additional equilibria below one. Since G^C is strictly increasing and $G^C(1) = G^{NC}(1)$, full acquisition gives equal accuracy across games, while every additional equilibrium of game C gives strictly lower accuracy. Hence $G^{*C} \leq G^{*NC}$.

It remains to consider $c \in (\Delta(1), \hat{c})$. Parametrize equilibria by $\alpha := \alpha^{*NC} \in (\hat{\alpha}, 1)$, and let

$\beta(\alpha) := \alpha^{*C}$ denote the solution on the decreasing branch of Δ^C satisfying

$$\Delta^C(\beta(\alpha)) = \Delta^{NC}(\alpha).$$

By Proposition 1, $\beta(\alpha) < \alpha$. Define

$$\widetilde{D}(\alpha) := G^C(\beta(\alpha)) - G^{NC}(\alpha).$$

Using (11) and the equilibrium condition,

$$\widetilde{D}'(\alpha) = N\Delta^{NC}(\alpha)\big(\beta'(\alpha) - 1\big). \tag{16}$$

At $\alpha = \hat{\alpha}$, $\beta(\hat{\alpha}) = \hat{\alpha}$, so the strict part of (3) gives $\widetilde{D}(\hat{\alpha}) > 0$. As $\alpha \to 1$, $\beta(\alpha) \to \alpha'$, where $\Delta^C(\alpha') = \Delta(1)$ on the decreasing branch. Since $\alpha' < 1$, G^C is strictly increasing, and $G^C(1) = G^{NC}(1)$,

$$\widetilde{D}(1^-) = G^C(\alpha') - G^C(1) < 0.$$

Thus $\widetilde{D}$ has at least one zero on $(\hat{\alpha}, 1)$.

To establish uniqueness, Lemma 5 implies that β' is strictly increasing. Moreover, the strict-convexity argument in the proof of Lemma 2 implies

$$(\Delta^C - \Delta^{NC})'(\hat{\alpha}) < 0.$$

Since both derivatives are negative at $\hat{\alpha}$,

$$\beta'(\hat{\alpha}) = \frac{\Delta^{NC\prime}(\hat{\alpha})}{\Delta^{C\prime}(\hat{\alpha})} < 1.$$

If $\beta'(\alpha) < 1$ throughout $(\hat{\alpha}, 1)$, (16) implies that $\widetilde{D}$ is strictly decreasing and therefore has exactly one zero. Otherwise, since β' is strictly increasing, there is a unique $\alpha^\dagger$ such that $\beta'(\alpha^\dagger) = 1$. Then $\widetilde{D}$ is strictly decreasing before $\alpha^\dagger$ and strictly increasing thereafter. Because $\widetilde{D}(1^-) < 0$, it remains negative on $[\alpha^\dagger, 1)$ and hence crosses zero exactly once before $\alpha^\dagger$.

Let α_G denote this unique zero and define

$$\hat{c}_G := \Delta^{NC}(\alpha_G).$$

Since $\alpha_G \in (\hat{\alpha}, 1)$ and Δ^{NC} is strictly decreasing, $\hat{c}_G \in (\Delta(1), \hat{c})$. Moreover, $\widetilde{D}(\alpha) > 0$ for

$\alpha < \alpha_G$ and $\widetilde{D}(\alpha) < 0$ for $\alpha > \alpha_G$. Since $c = \Delta^{NC}(\alpha)$ is strictly decreasing in α,

$$G^{*C} > G^{*NC} \quad \text{for } c \in (\hat{c}_G, \hat{c}), \qquad G^{*C} < G^{*NC} \quad \text{for } c \in (\Delta(1), \hat{c}_G).$$

Together with the two regions above, this proves the proposition. □

B.4 Proof for Welfare

Social-optimum benchmark. Let $\alpha^{X,FB}$ denote any welfare-maximizing acquisition rate in game X. By (11),

$$W^{X\prime}(\alpha) = N\Delta^X(\alpha) - c.$$

Thus, any interior welfare optimum satisfies $\Delta^X(\alpha^{X,FB}) = c/N$, whereas any interior equilibrium satisfies $\Delta^X(\alpha^{*X}) = c$. This is the basic wedge between the social and private values of information acquisition.

For game NC, Δ^{NC} is strictly decreasing, so W^{NC} is strictly concave. If $c \in (\Delta(1), \Delta^{NC}(0))$, the equilibrium is interior and

$$W^{NC\prime}(\alpha^{*NC}) = (N-1)c > 0,$$

so $\alpha^{NC,FB} > \alpha^{*NC}$. If $c \in [\Delta^{NC}(0), \Delta^C(0))$, then $\alpha^{*NC} = 0$. Since $\binom{2n}{n}$ is strictly larger than the average coefficient in the expansion of $(1+1)^{2n}$,

$$N\binom{2n}{n} > 4^n,$$

and hence $N\Delta^{NC}(0) > \Delta^C(0) > c$. Thus $W^{NC\prime}(0) > 0$ and again $\alpha^{NC,FB} > \alpha^{*NC}$. If $c \geq \Delta^C(0)$, $\alpha^{*NC} = 0 \leq \alpha^{NC,FB}$ trivially. Finally, if $c \leq \Delta(1)$, then $\Delta^{NC}(\alpha) \geq \Delta(1)$ for all α, so $W^{NC\prime}(\alpha) \geq N\Delta(1) - c > 0$ and therefore $\alpha^{NC,FB} = \alpha^{*NC} = 1$.

For game C, if $c \geq \Delta^C(0)$, then $\alpha^{*C} = 0 \leq \alpha^{C,FB}$. If $c \in (\Delta(1), \Delta^C(0))$, the unique equilibrium lies on the decreasing branch of Δ^C. Hence $\Delta^C(t) \geq c$ for all $t \leq \alpha^{*C}$ and $W^{C\prime}(\alpha^{*C}) = (N-1)c > 0$. Welfare is therefore increasing up to α^{*C} and continues to increase immediately to its right, so every welfare-maximizing acquisition rate satisfies $\alpha^{C,FB} > \alpha^{*C}$.

Finally, suppose $c \leq \Delta(1)$. The $k = n$ term in (5) gives $\Delta^C(t) \geq \Delta(1)t^{2n}$, with strict

inequality for $t < 1$ on a set of positive measure. Hence, for every $a \in [0, 1)$,

$$\begin{aligned} W^C(1) - W^C(a) &= N \int_a^1 \Delta^C(t)\, dt - (1-a)c \\ &> \Delta(1)(1 - a^N) - (1-a)c \\ &\geq \Delta(1)(a - a^N) \geq 0. \end{aligned}$$

Thus $\alpha^{C,FB} = 1$, so every equilibrium acquisition rate in game C is weakly below the welfare-maximizing rate.

Therefore, equilibrium information acquisition is weakly below the welfare-maximizing level in both games, and strictly below it for intermediate information costs.

Proof of Proposition 3. Recall the function β from the proof of Proposition 2: for $\alpha \in (\hat{\alpha}, 1)$, $\beta(\alpha)$ is the equilibrium acquisition rate in game C corresponding to the equilibrium rate α in game NC, with

$$\Delta^C(\beta(\alpha)) = \Delta^{NC}(\alpha), \qquad \beta(\alpha) < \alpha.$$

Also recall that β is strictly convex and β' is strictly increasing.

We first establish the welfare comparison outside the region where the cutoff can arise. Suppose $c \in [\hat{c}_G, \hat{c}]$. By Proposition 2, $G^{*C} \geq G^{*NC}$, while Proposition 1 gives $\alpha^{*C} \leq \alpha^{*NC}$. Hence

$$W^{*C} - W^{*NC} = (G^{*C} - G^{*NC}) + c(\alpha^{*NC} - \alpha^{*C}) \geq 0.$$

The inequality is strict throughout $[\hat{c}_G, \hat{c}]$: at $c = \hat{c}_G$ the accuracy terms are equal but $\alpha^{*C} < \alpha^{*NC}$, while at $c = \hat{c}$ the acquisition rates are equal but $G^{*C} > G^{*NC}$.

If $c \in (\hat{c}, \Delta^C(0))$, Proposition 1 gives $0 \leq \alpha^{*NC} < \alpha^{*C} < \hat{\alpha}$. Using (3) and (12),

$$\begin{aligned} W^{*C} - W^{*NC} &= \left[G^C(\alpha^{*NC}) - G^{NC}(\alpha^{*NC})\right] + N \int_{\alpha^{*NC}}^{\alpha^{*C}} \Delta^C(t)\, dt - c(\alpha^{*C} - \alpha^{*NC}) \\ &\geq (N-1)c(\alpha^{*C} - \alpha^{*NC}) > 0, \end{aligned}$$

where $\Delta^C(t) \geq \Delta^C(\alpha^{*C}) = c$ on the relevant range. If $c \geq \Delta^C(0)$, both games have $\alpha^* = 0$, so $W^{*C} = W^{*NC} = 1/2$. Thus $W^{*C} \geq W^{*NC}$ for all $c \geq \hat{c}_G$, with strict inequality for $c \in [\hat{c}_G, \Delta^C(0))$.

If $c \leq \Delta(1)$, game NC has the unique equilibrium $\alpha^{*NC} = 1$, while game C admits $\alpha^{*C} = 1$ and may admit additional equilibria below one. By the social-optimum benchmark above, $W^C(1) > W^C(a)$ for every $a < 1$, and $W^C(1) = W^{NC}(1)$. Hence $W^{*C} \leq W^{*NC}$ in this region, with equality when $\alpha^{*C} = 1$.

It remains to consider $c \in (\Delta(1), \hat{c}_G)$. Parametrize equilibria by $\alpha := \alpha^{*NC} \in (\alpha_G, 1)$, where α_G is the unique zero from the proof of Proposition 2, and let $\beta(\alpha) = \alpha^{*C}$. Define

$$\widetilde{W}(\alpha) := W^C(\beta(\alpha)) - W^{NC}(\alpha) = \widetilde{D}(\alpha) + (\alpha - \beta(\alpha))\Delta^{NC}(\alpha),$$

where $\widetilde{D}(\alpha) := G^C(\beta(\alpha)) - G^{NC}(\alpha)$.
At $\alpha = \alpha_G$, $\widetilde{D}(\alpha_G) = 0$ and $\beta(\alpha_G) < \alpha_G$, so

$$\widetilde{W}(\alpha_G) = (\alpha_G - \beta(\alpha_G))\Delta^{NC}(\alpha_G) > 0.$$

As $\alpha \to 1^-$, $\beta(\alpha) \to \alpha' < 1$, where $\Delta^C(\alpha') = \Delta(1) = \Delta^C(1)$. Thus, evaluating welfare at $c = \Delta(1)$,

$$\widetilde{W}(1^-) = W^C(\alpha') - W^C(1) < 0$$

by the social-optimum benchmark above. Hence $\widetilde{W}$ has at least one zero on $(\alpha_G, 1)$.
We next show that this zero is unique. Write $c = c(\alpha) := \Delta^{NC}(\alpha)$. Differentiating $\widetilde{W}$ and using (11) together with $\Delta^C(\beta) = \Delta^{NC}(\alpha) = c$ gives

$$\widetilde{W}'(\alpha) = (N-1)c(\beta'(\alpha) - 1) + c'(\alpha - \beta(\alpha)). \tag{17}$$

Since $c' < 0$ and $\alpha - \beta > 0$, $\widetilde{W}'(\alpha) < 0$ whenever $\beta'(\alpha) \leq 1$. The proof of Proposition 2 implies $\beta'(\alpha_G) < 1$, while Lemma 5 implies that β' is strictly increasing.
At any critical point $\alpha_m \in (\alpha_G, 1)$, (17) implies $\beta'_m > 1$ and

$$(N-1)c_m(\beta'_m - 1) = |c'_m|(\alpha_m - \beta_m). \tag{18}$$

Differentiating (17) and using (18) gives

$$\widetilde{W}''(\alpha_m) = (N-1)c_m\beta''_m + (\alpha_m - \beta_m)\left[c''_m - \frac{N-2}{N-1}\frac{(c'_m)^2}{c_m}\right]. \tag{19}$$

Let $d := q - \frac{1}{2}$, $f := \Delta^C/d$, and $g := \Delta^{NC}/d$, so $c = dg$ and $f(\beta) = g(\alpha)$. Recall from the proof of Lemma 5 that

$$P := (2n-1)g'^2 - 2ngg'' > 0, \qquad \frac{f''(\beta)}{f'(\beta)^2} \geq \frac{2n-1}{2n\, g(\alpha)}.$$

Since $N - 1 = 2n$ and $N - 2 = 2n - 1$,

$$c'' - \frac{N-2}{N-1}\frac{c'^2}{c} = -\frac{dP}{2ng}, \qquad \beta'' \geq \frac{P\beta'}{2ng\,|g'|}.$$

At the critical point, (18) also gives

$$\alpha_m - \beta_m = \frac{2ng(\beta'_m - 1)}{|g'|}.$$

Substituting into (19),

$$\widetilde{W}''(\alpha_m) \geq \frac{dP}{|g'|}\big[\beta'_m - (\beta'_m - 1)\big] = \frac{dP}{|g'|} > 0.$$

Thus every critical point of $\widetilde{W}$ on $(\alpha_G, 1)$ is a strict local minimum.
It follows that $\widetilde{W}'$ has at most one zero. Indeed, $\widetilde{W}' < 0$ initially, and at any zero of $\widetilde{W}'$ we have $\widetilde{W}'' > 0$, so $\widetilde{W}'$ can cross zero only from negative to positive. Hence $\widetilde{W}$ is either strictly decreasing or first decreasing and then increasing. Since $\widetilde{W}(\alpha_G) > 0$ and $\widetilde{W}(1^-) < 0$, it crosses zero exactly once.
Let $\alpha_W \in (\alpha_G, 1)$ denote this unique zero and define

$$\hat{c}_W := \Delta^{NC}(\alpha_W).$$

Since Δ^{NC} is strictly decreasing and $\alpha_W > \alpha_G$, $\hat{c}_W \in (\Delta(1), \hat{c}_G)$. Moreover, $\widetilde{W}(\alpha) > 0$ for $\alpha \in (\alpha_G, \alpha_W)$ and $\widetilde{W}(\alpha) < 0$ for $\alpha \in (\alpha_W, 1)$. Since $c = \Delta^{NC}(\alpha)$ is strictly decreasing in α, this implies

$$W^{*C} > W^{*NC} \quad \text{for } c \in (\hat{c}_W, \hat{c}_G), \qquad W^{*C} < W^{*NC} \quad \text{for } c \in (\Delta(1), \hat{c}_W).$$

Combining these comparisons with the regions established above proves the proposition. □

C Robustness and Extensions

C.1 Robustness

This subsection relaxes several baseline assumptions one at a time in the three-voter case, where each relaxation admits a sharp characterization. In the baseline model with $N = 3$, communication raises the value of information acquisition when $\alpha < 1/3$ and lowers it when $\alpha \in (1/3, 1)$. Together with the fact that Δ^{NC} is strictly decreasing in α, this single-crossing property delivers the equilibrium comparison in Proposition 1. Each part below therefore first shows that the two-sided effect on the value of acquisition survives and then discusses its implications for equilibrium acquisition.

C.1.1 Unbalanced prior

Suppose the common prior is unbalanced. Without loss of generality, let

$$\pi := \Pr(z = A) \in \left(\frac{1}{2}, q\right),$$

so that A is ex ante more likely, while a signal favoring B is sufficiently informative to overturn the prior. I retain symmetry across voters but no longer impose neutrality across alternatives, since relabeling A and B changes the prior.

In game C, the full-revelation argument in Lemma 1 continues to apply. In game NC, by contrast, being pivotal can reveal information about the state, so voting must be recharacterized.

Fix an acquisition probability α. Solving the symmetric voting subgame in game NC, a voter who acquires follows his or her signal, while an uninformed voter votes for A with probability λ. Either of the other two voters votes for A with probability

$$\rho_A = \alpha q + (1-\alpha)\lambda, \qquad \rho_B = \alpha(1-q) + (1-\alpha)\lambda$$

in states A and B, respectively. An uninformed voter is indifferent when pivotal if

$$\pi\rho_A(1-\rho_A) = (1-\pi)\rho_B(1-\rho_B). \tag{20}$$

Define

$$\bar{\alpha} := \frac{q-\pi}{(1-\pi)q^2 - \pi(1-q)^2} \in (0,1).$$

The equilibrium voting probability is

$$\lambda(\alpha) = \begin{cases} 1, & \alpha = 0 \text{ or } \alpha \geq \bar{\alpha}, \\ \lambda_\alpha \in (0,1), & 0 < \alpha < \bar{\alpha}, \end{cases}$$

where λ_α is the unique solution to (20). At $\alpha = 0$, $\lambda = 1$ is selected by the weakly undominated voting restriction. Moreover, $\lambda_\alpha \to 1$ as $\alpha \downarrow 0$.

Let $\Delta_\pi^X(\alpha)$ denote the value of acquisition under prior π. At $\alpha = 0$, both other voters in game NC vote for A, so a unilateral acquisition cannot change the collective decision, whereas in game C it allows the acquired signal to determine the decision. Hence

$$\Delta_\pi^{NC}(0) = 0, \qquad \Delta_\pi^{C}(0) = q - \pi > 0.$$

Since $\lambda_\alpha \to 1$ as $\alpha \downarrow 0$, this strict inequality persists for all sufficiently small α.

For $\alpha \geq \bar{\alpha}$, uninformed voters in game NC vote for A with probability one. Let δ_k^X denote the value of acquisition conditional on exactly $k \in \{0, 1, 2\}$ of the other two voters acquiring. Direct calculation gives

	$k=0$	$k=1$	$k=2$
δ_k^{NC}	0	$(1-\pi)q^2 - \pi(1-q)^2$	$2q(1-q)(q-\pi)$
δ_k^{C}	$q-\pi$	$(2\pi-1)q(1-q)$	$2q(1-q)(q-\pi)$.

Thus

$$\delta_0^C - \delta_0^{NC} = q - \pi, \qquad \delta_1^C - \delta_1^{NC} = -(q-\pi), \qquad \delta_2^C - \delta_2^{NC} = 0,$$

and therefore, for $\alpha \geq \bar{\alpha}$,

$$\Delta_\pi^C(\alpha) - \Delta_\pi^{NC}(\alpha) = (q-\pi)(1-\alpha)(1-3\alpha).$$

Consequently, communication strictly raises the value of acquisition for all sufficiently small α, while it strictly lowers the value for

$$\alpha \in \left(\max\left\{ \bar{\alpha}, \frac{1}{3} \right\}, 1 \right).$$

Thus the two-sided persuasion–substitution effect survives an unbalanced prior, although the voting behavior of uninformed voters in game NC is no longer neutral.

Unlike in the baseline model, however, game NC admits the no-acquisition equilibrium at every cost: when no other voter acquires, uninformed voters all vote for the ex ante favored alternative, so a single acquirer is never pivotal and $\Delta_\pi^{NC}(0) = 0$. Equilibrium comparisons therefore depend on equilibrium selection in game NC.

C.1.2 Informative signals for non-acquirers

Suppose every voter receives a private signal. Without acquiring, the signal has precision $p \in (1/2, q)$; acquiring upgrades its precision to q. Assume that the message space in game C is rich enough to distinguish the four post-acquisition types. Because the prior remains balanced, the post-acquisition characterization in Lemma 1 extends directly: in game NC, voters vote sincerely according to their private signals, while in game C, communication fully reveals signal precisions and realizations and voting is sincere.

Assume

$$q > \frac{p^2}{p^2 + (1-p)^2},$$

so that one high-precision signal outweighs two concordant low-precision signals pointing the other way.[37]

Let δ_k^X denote the value of acquisition in game X conditional on exactly $k \in \{0,1,2\}$ of the other two voters acquiring. Enumerating the three cases gives

	$k=0$	$k=1$	$k=2$
δ_k^{NC}	$2p(1-p)(q-p)$	$(q-p)(p+q-2pq)$	$2q(1-q)(q-p)$
δ_k^{C}	$q-(3p^2-2p^3)$	$q(1-q)(2p-1)$	$2q(1-q)(q-p).$

Thus,

$$\delta_0^C - \delta_0^{NC} = A(p,q), \qquad \delta_1^C - \delta_1^{NC} = -A(p,q), \qquad \delta_2^C - \delta_2^{NC} = 0,$$

where $A(p,q) := q[p^2 + (1-p)^2] - p^2 > 0$. Averaging over the number of other acquirers,

$$\Delta_{p,q}^C(\alpha) - \Delta_{p,q}^{NC}(\alpha) = A(p,q)(1-\alpha)(1-3\alpha).$$

Hence communication raises the value of acquisition for $\alpha < 1/3$ and lowers it for $\alpha \in (1/3, 1)$. Allowing non-acquirers to hold informative but less precise signals therefore leaves the baseline single-crossing pattern unchanged. Moreover, $\delta_0^{NC} - \delta_1^{NC} = (q-p)^2(2p-1) > 0$ and $\delta_1^{NC} - \delta_2^{NC} = (q-p)^2(2q-1) > 0$, so $\Delta_{p,q}^{NC}$ is strictly decreasing in α, and the equilibrium comparison in Proposition 1 extends to this environment.

C.1.3 Asymmetric acquisition equilibria

The baseline model restricts attention to symmetric acquisition strategies. Here I drop this restriction at the acquisition stage and consider pure-strategy equilibria for $N = 3$.

Let $d := q - \frac{1}{2}$ and $r := q(1-q)$, and let δ_k^X denote the value of acquisition in game X when exactly $k \in \{0,1,2\}$ of the other two voters acquire. Under the baseline information structure,

	$k=0$	$k=1$	$k=2$
δ_k^{NC}	$\frac{d}{2}$	$\frac{d}{2}$	$2dr$
δ_k^{C}	d	0	$2dr$

with $2dr < d/2 < d$. Thus communication raises the value of acquisition when neither other voter acquires, lowers it when exactly one acquires, and has no effect when both acquire.

A pure profile with exactly m acquirers is an equilibrium if no acquirer wants to stop

[37] Under the reverse strict inequality, two concordant low-precision signals outweigh a conflicting high-precision signal. The two games then use the available signals in the same way, so communication does not change the value of acquisition.

and no non-acquirer wants to acquire, i.e.,

$$\delta^X_{m-1} \geq c \quad \text{and} \quad \delta^X_m \leq c,$$

whenever the corresponding condition applies. Away from the cutoff values, the equilibrium numbers of acquirers are therefore

Cost range	NC	C
$c > d$	0	0
$\frac{d}{2} < c < d$	0	1
$2dr < c < \frac{d}{2}$	2	1
$0 < c < 2dr$	3	1 or 3

At the cutoff values, additional equilibrium configurations may arise because some voters are indifferent.

Hence, over the higher intermediate-cost range communication increases acquisition from zero to one acquirer, whereas over the lower intermediate-cost range it reduces acquisition from two acquirers to one. When information is sufficiently cheap, game NC has three acquirers while game C may have either one or three. The cost-dependent persuasion–substitution pattern therefore does not rely on symmetric mixing. Away from the cutoff costs, communication increases equilibrium acquisition at higher intermediate costs and reduces it at lower intermediate costs, as in Proposition 1, although the switching point is $d/2$ rather than $\hat{c}$. For the experimental parameterization $q = 0.9$, we have $2dr = 0.072$, $d/2 = 0.20$, and $d = 0.40$. Thus the low cost $c = 0.10$ implies two acquirers in game NC and one in game C, whereas the high cost $c = 0.25$ implies zero acquirers in game NC and one in game C.

C.1.4 Imperfect information transmission

The baseline model assumes full information transmission in game C. As a reduced-form relaxation, suppose that after acquisition each voter's post-acquisition type is publicly disclosed with probability $\rho \in [0, 1]$, independently across voters. An undisclosed type remains private to the voter. Thus $\rho = 0$ is outcome-equivalent to game NC, while $\rho = 1$ gives the full-revelation case.

Because some private information may remain undisclosed, the post-acquisition argument in Lemma 1 does not apply directly. I select the symmetric and neutral weakly undominated continuation in which voters vote sincerely, with an informed voter following his

or her own signal when the posterior is tied and an uninformed voter randomizing at an uninformative tie.[38]

Let $d := q - \frac{1}{2}$, $r := q(1-q)$, and let δ_k^ρ denote the value of acquisition when exactly $k \in \{0,1,2\}$ of the other two voters acquire. Under this continuation,

	$k=0$	$k=1$	$k=2$
δ_k^ρ	$\frac{d}{2}(1+\rho)$	$\frac{d}{2}(1-\rho)$	$2dr.$

At $\rho = 0$ this reduces to game NC, while at $\rho = 1$ it reduces to game C. Hence

$$\delta_0^\rho - \delta_0^{NC} = \frac{\rho d}{2}, \qquad \delta_1^\rho - \delta_1^{NC} = -\frac{\rho d}{2}, \qquad \delta_2^\rho - \delta_2^{NC} = 0.$$

Averaging over the number of other acquirers gives

$$\Delta^\rho(\alpha) - \Delta^{NC}(\alpha) = \frac{\rho d}{2}(1-\alpha)(1-3\alpha).$$

Therefore, for every $\rho > 0$, partial information transmission raises the value of acquisition when $\alpha < 1/3$ and lowers it when $\alpha \in (1/3, 1)$. Imperfect transmission thus attenuates the effect of communication but leaves the persuasion–substitution cutoff unchanged. Since game NC is unaffected by ρ and the cutoff remains $\hat{\alpha} = 1/3$, the cost cutoff $\hat{c} = \Delta^{NC}(1/3)$ is also unchanged. The equilibrium comparison in Proposition 1 therefore continues to hold in its weak form, while the strict comparisons now arise for $c \in (\Delta(1), \Delta^\rho(0))$, where $\Delta^\rho(0) = \frac{d}{2}(1+\rho) < \Delta^C(0)$ for $\rho < 1$.

C.2 Abstention

Suppose voters may abstain, with majority rule applied to votes cast and ties broken uniformly at random. The full-revelation continuation of game C remains unchanged. In game NC, however, abstention creates additional continuation equilibria.

Proposition 4 (Abstention)**.** *Fix a symmetric acquisition probability α. Game NC admits both of the following symmetric and neutral continuation equilibria in weakly undominated voting strategies:*

1. *informed voters vote according to their signals and non-acquirers abstain; the value of acquisition is $\Delta^C(\alpha)$;*

[38] With neither other type disclosed, neutrality makes pivotality uninformative, as in game NC; with both disclosed, no private information remains. With exactly one other type disclosed, pivotality can be informative but never reverses a strict posterior ranking. When the voter's private signal exactly offsets a disclosed signal, pivotality favors the private-signal tie break.

2. *informed voters vote according to their signals and non-acquirers randomize uniformly between A and B; the value of acquisition is $\Delta^{NC}(\alpha)$.*

Hence allowing abstention can either eliminate the communication wedge in acquisition incentives or leave the baseline two-sided effect unchanged.

Proof. For the first continuation, suppose all other voters vote according to their signals if informed and abstain otherwise. A voter can affect the outcome only when the votes cast by the others are tied or differ by one. A tie is uninformative about the state. If one alternative leads by one vote, that alternative is more likely to be correct: conditional on any odd number of acquired signals, the likelihood ratio of a one-vote lead for the correct rather than the incorrect alternative is $q/(1-q) > 1$. Hence abstention is a best response for a non-acquirer, and a strict one when $0 < \alpha < 1$. For an informed voter, a tie among the other votes leaves her signal decisive; when the others favor the opposite alternative by one vote, the likelihood ratio of that lead exactly offsets her signal, leaving her indifferent between voting her signal and abstaining. Voting according to her signal is therefore a best response.

Under this continuation, only acquired signals are counted. The majority decision therefore follows the sign of the realized signal imbalance, with a random decision when the signals are tied. This is exactly the decision rule induced by full revelation in game C, so the value of acquisition is $\Delta^{C}(\alpha)$.

For the second continuation, all other $2n$ voters cast a vote, so their vote margin is even and voter i can affect the outcome only when the other votes are tied. By neutrality, this event is uninformative about the state. An uninformed voter is thus indifferent between A, B, and abstention, while an informed voter strictly prefers to follow her signal whenever the tie event occurs. This continuation therefore reproduces the baseline post-acquisition outcome in game NC, so the value of acquisition is $\Delta^{NC}(\alpha)$.

It remains to verify weak undominance. Voting according to one's signal is strictly optimal when all other voters abstain. Abstention is strictly optimal for a non-acquirer when one other voter votes according to an informative signal and all remaining voters abstain, so it is not weakly dominated. Now suppose a mixed action $(x_A, x_B, x_\varnothing)$ weakly dominates uniform randomization between A and B. Consider an opponent profile under which voter i is pivotal only when an informative voter votes for A. Weak dominance requires $x_A \geq x_B$; the symmetric profile requires $x_B \geq x_A$, so $x_A = x_B$. Finally, if one other voter votes against an informative signal and all remaining voters abstain, uniform voting is strictly better than abstention. Hence $x_\varnothing = 0$, implying $x_A = x_B = 1/2$. Thus uniform randomization is also weakly undominated. □

C.3 Large groups

Proposition 5 (Large groups)**.** *For any $c \in (0, q - \frac{1}{2})$, there exists $\bar{n}$ such that, for all $n \geq \bar{n}$, game NC has the unique equilibrium $\alpha^{*NC} = 0$, while game C has the unique interior equilibrium $\alpha^{*C} \in (0, 1)$. Moreover, $G^{*C} > \frac{1}{2} = G^{*NC}$ and $W^{*C} > \frac{1}{2} = W^{*NC}$.*

Proof. Write $d := q - \frac{1}{2} > 0$. In game NC, conditional on either state, each other voter votes for the correct alternative with probability $\frac{1}{2} + \alpha d$ and for the incorrect alternative with probability $\frac{1}{2} - \alpha d$. Hence

$$\Delta^{NC}(\alpha) = d\binom{2n}{n}\left(\frac{1}{4} - \alpha^2 d^2\right)^n \leq d\binom{2n}{n}4^{-n}.$$

Since $\binom{2n}{n}4^{-n} \to 0$, the right-hand side converges to zero uniformly in α. Thus, for all sufficiently large n, $\Delta^{NC}(\alpha) < c$ for every $\alpha \in [0, 1]$, so $\alpha^{*NC} = 0$ is the unique equilibrium. For game C, note that

$$\Delta^{C}(0) = d > c,$$

while

$$\Delta^{C}(1) = \Delta^{NC}(1) = d\binom{2n}{n}[q(1-q)]^n \longrightarrow 0,$$

because $q(1 - q) < 1/4$. Hence, for all sufficiently large n,

$$\Delta^{C}(1) < c < \Delta^{C}(0).$$

Lemma 3 therefore implies that game C has a unique interior equilibrium $\alpha^{*C} \in (0, 1)$. In particular, $\Delta^{C}(\alpha^{*C}) = c$, so the probability of information-pivotal events at the equilibrium of game C equals c/d for all sufficiently large n.

Since $\alpha^{*NC} = 0$, we have

$$G^{*NC} = W^{*NC} = \frac{1}{2}.$$

Because $\alpha^{*C} > 0$ and $G^{C}(\alpha)$ is strictly increasing,

$$G^{*C} = G^{C}(\alpha^{*C}) > \frac{1}{2}.$$

Finally, equation (11) gives

$$G^{C}(\alpha) = \frac{1}{2} + N\int_0^{\alpha} \Delta^{C}(t)\, dt.$$

Since α^{*C} is the unique solution to $\Delta^{C}(\alpha) = c$ and $\Delta^{C}(0) > c$, continuity implies $\Delta^{C}(t) > c$

for all $t \in [0, \alpha^{*C})$. Therefore,

$$W^{*C} - \frac{1}{2} = N \int_0^{\alpha^{*C}} \Delta^C(t)\, dt - \alpha^{*C} c$$
$$> (N-1)\alpha^{*C} c > 0.$$

Thus $W^{*C} > \frac{1}{2} = W^{*NC}$. □

D Supplementary Experimental Results

D.1 Additional predictions

Table 5: Symmetric equilibrium predictions under the experimental parameterization

	Low cost ($c = 0.10$)			High cost ($c = 0.25$)		
	Acquisition	Accuracy	Welfare	Acquisition	Accuracy	Welfare
No communication	0.883	0.942	0.854	0	0.500	0.500
Communication	0.560	0.879	0.823	0.215	0.707	0.653

Notes: Panel B reports the symmetric acquisition rate that maximizes welfare $W^X(\alpha) = G^X(\alpha) - \alpha c$ in each game, together with the resulting group accuracy and welfare.

D.2 Individual characteristics

After the main task, subjects completed three incentivized tasks measuring risk attitude, attitude toward strategic uncertainty, and social preferences. One of the three tasks was randomly selected for payment.

Risk attitude. Following Gneezy and Potters (1997), subjects received 100 points and chose how many to invest in a risky option. Invested points were lost with probability one half and multiplied by 2.5 with probability one half, while uninvested points were retained for sure. I use the number of points invested as the measure of risk tolerance.

Attitude toward strategic uncertainty. Following Heinemann et al. (2009) and Alós-Ferrer and Garagnani (2020), subjects were matched in groups of three and made ten choices between a safe option and a coordination-dependent option. The safe payoff increased across decisions, while the coordination-dependent option paid 100 points if at least two group

members chose it and zero otherwise. I use the last decision in which a subject chose the coordination-dependent option as the measure of willingness to bear strategic uncertainty.

Social preferences. Social preferences were measured using the six-item Social Value Orientation (SVO) Slider Measure of Murphy et al. (2011). In each item, subjects chose an allocation of points between themselves and another participant. I use the resulting SVO angle, with a larger value indicating a more prosocial orientation.

Table 6 reports these measures, together with gender and age, across treatments. None differs significantly across the four treatments.

Table 6: Individual characteristics across treatments

	Low cost		High cost		
	No comm.	Comm.	No comm.	Comm.	p-value
Risk investment	50.96	53.15	54.55	56.18	0.628
Strategic uncertainty	5.81	6.24	6.27	5.78	0.564
SVO angle	21.09	20.95	22.33	24.12	0.424
Female	0.50	0.61	0.61	0.60	0.190
Age	22.25	22.08	22.79	22.67	0.799

Notes: Entries are means of matching-group averages. Female is an indicator for reporting being female. The p-value is from a Kruskal–Wallis test across the four treatments. The matching group is the unit of randomization and independent observation.

D.3 Results including the excluded matching group

Table 7 reports the high-cost treatment comparisons when the excluded matching group is retained. The estimated effects remain positive for all three outcomes. The acquisition effect becomes smaller and is no longer statistically significant, while the effects on group accuracy and welfare remain significant.

Table 7: Robustness including the excluded matching group (c=0.25)

Sample	Outcome	No comm.	Comm.	Difference	p-value
Main sample	Information acquisition	0.308	0.384	0.076	0.042
	Group accuracy	0.703	0.796	0.093	0.015
	Welfare	0.626	0.700	0.074	0.029
Full sample	Information acquisition	0.341	0.384	0.043	0.102
	Group accuracy	0.714	0.796	0.082	0.032
	Welfare	0.629	0.700	0.071	0.023

Notes: The table reports high-cost treatment means at the matching-group level. The main sample excludes one no-communication matching group because of the procedural error described in Section 4; the full sample retains it. The p-value for information acquisition is from the preregistered one-sided Mann–Whitney test; p-values for group accuracy and welfare are from two-sided Mann–Whitney tests.

D.4 Additional results on information acquisition

Figure 7 shows acquisition rates across rounds. The communication–no-communication differences persist over much of the experiment rather than being driven by a small number of rounds. Figure 8 shows substantial heterogeneity in individual acquisition frequencies within each treatment. The distribution under communication tends to shift toward more frequent acquisition, particularly under low information cost.

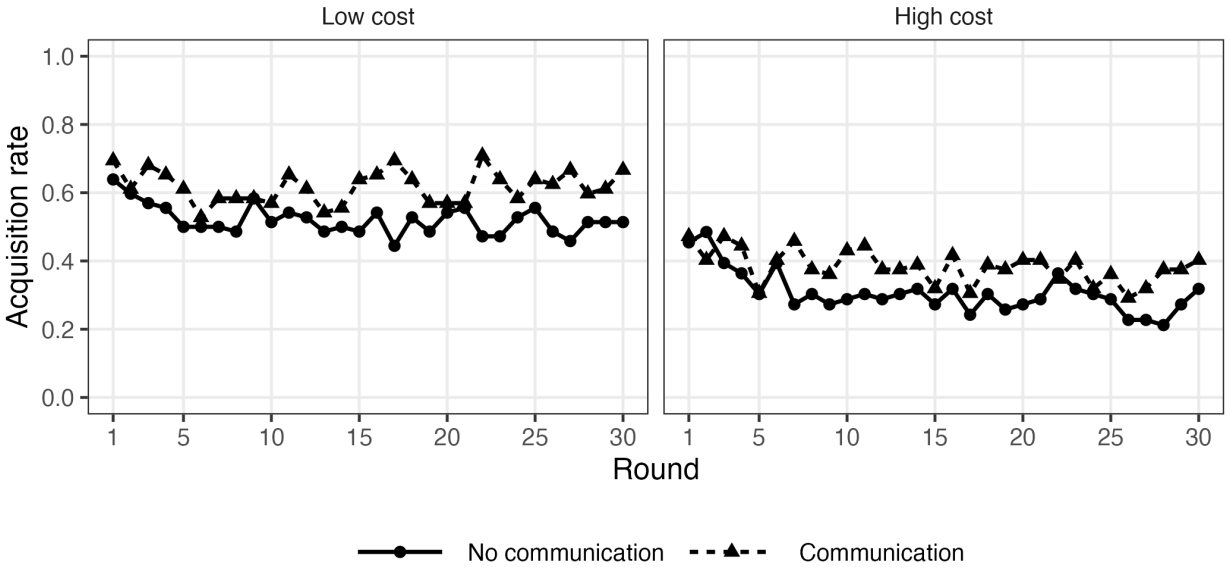


Figure 7: Information acquisition by round

Table 8 reports additional regression analyses. The pooled and cost-specific communication effects are stable to adding individual characteristics, excluding round 30, and restricting the sample to rounds 16–30. Including the previously excluded matching group attenuates the pooled and high-cost effects, consistent with the nonparametric comparisons in Section D.3. Among the individual characteristics, a higher SVO angle is positively associated with information acquisition, while the other characteristics show no statistically significant association.

D.5 Additional results on messages and voting

Table 9 reports message choices conditional on post-acquisition type. Messaging behavior is similar across cost conditions. As discussed in Section 4, subjects who acquire information almost always report their realized signal truthfully, whereas subjects who do not acquire usually send no message but sometimes claim to have observed a signal.

Table 8: Additional analyses of information acquisition

	(1) Baseline	(2) Characteristics	(3) Excl. R30	(4) Rounds 16–30	(5) Full sample
Panel A. Communication effects					
Pooled	0.087**	0.083**	0.086**	0.104***	0.070*
	(0.035)	(0.031)	(0.035)	(0.038)	(0.038)
Low cost	0.098*	0.099**	0.096*	0.121**	0.098*
	(0.052)	(0.046)	(0.053)	(0.056)	(0.052)
High cost	0.076*	0.067	0.076	0.085*	0.043
	(0.045)	(0.045)	(0.046)	(0.051)	(0.054)
H2: one-sided p-value	0.050	0.073	0.053	0.050	0.216
Panel B. Individual characteristics					
Risk investment		0.0004			
		(0.0007)			
Strategic uncertainty		0.0066			
		(0.0077)			
SVO angle		0.0062***			
		(0.0014)			
Female		-0.0389			
		(0.0442)			
Age		-0.0020			
		(0.0054)			
Round controls	Yes	Yes	Yes	Yes	Yes
Observations	8,460	8,460	8,178	4,230	8,640
Matching groups	47	47	47	47	48

Notes: OLS; the dependent variable is an indicator for information acquisition. Column (1) reproduces the pooled and cost-specific estimates from Columns (2) and (4), respectively, of Table 2. Pooled estimates constrain the communication effect to be common across costs; cost-specific estimates allow it to differ by cost. All regressions include a high-cost indicator, centered round, and Round × Communication. Column (2) adds the characteristics in Panel B; columns (3)–(5) respectively exclude round 30, use rounds 16–30, and include the excluded matching group. Standard errors are clustered by matching group. Stars use two-sided tests; H2 p-values are one-sided. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 9: Message choices by post-acquisition type and information cost

	Low cost			High cost		
Post-acquisition type	–	R	B	–	R	B
–	0.84 (693)	0.08 (66)	0.08 (67)	0.90 (1196)	0.06 (78)	0.04 (57)
R	0.00 (0)	1.00 (681)	0.00 (1)	0.01 (3)	0.99 (398)	0.00 (2)
B	0.00 (0)	0.00 (1)	1.00 (651)	0.01 (5)	0.00 (1)	0.99 (420)

Notes: Entries are message frequencies conditional on post-acquisition type; observation counts are in parentheses. R and B denote red and blue, respectively; "–" denotes no signal in the row labels and no message in the column labels.

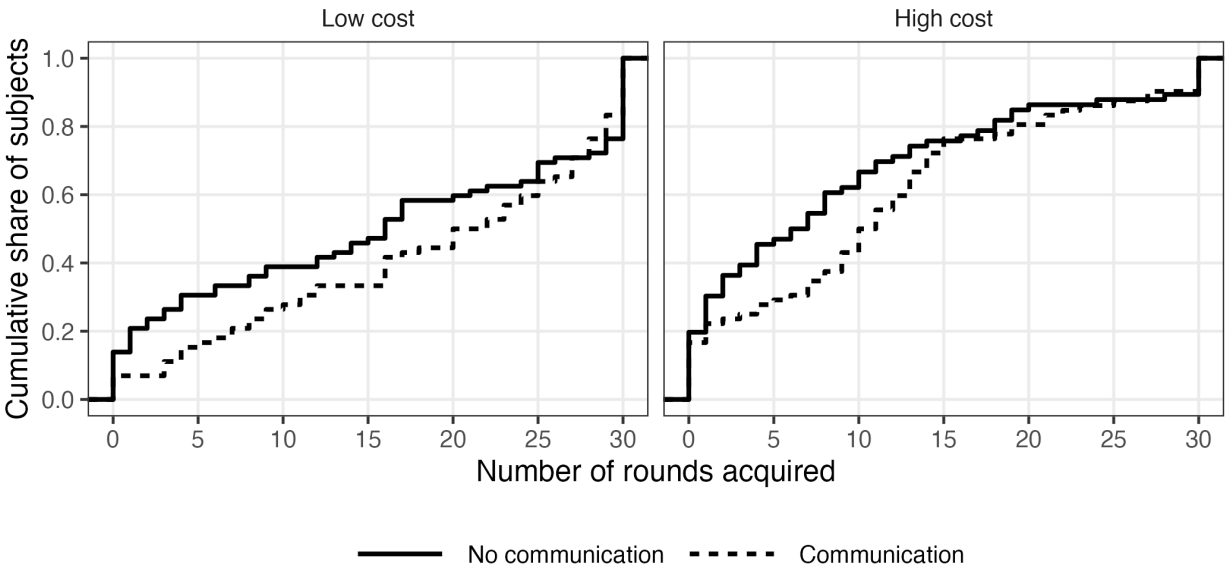


Figure 8: Distribution of individual information acquisition

Notes: The figure shows empirical cumulative distribution functions of the number of rounds in which each subject acquired information.

Table 10 reports the cost-specific voting frequencies underlying the pooled results in Table 3. Voting patterns are similar across the two cost conditions.

Table 10: Frequency of voting for Red by information cost

Panel A. Low cost							
	C: others' messages						NC
Own signal	B,B	B,–	R,B	–,–	R,–	R,R	
–	0.03 (135)	0.06 (198)	0.56 (86)	0.54 (72)	0.92 (176)	0.98 (159)	0.54 (1037)
R	0.05 (22)	0.82 (61)	1.00 (77)	0.97 (64)	1.00 (255)	1.00 (203)	0.99 (590)
B	0.00 (216)	0.00 (205)	0.00 (65)	0.04 (81)	0.26 (57)	0.96 (28)	0.00 (533)
Panel B. High cost							
	C: others' messages						NC
Own signal	B,B	B,–	R,B	–,–	R,–	R,R	
–	0.01 (101)	0.05 (331)	0.50 (66)	0.54 (375)	0.96 (363)	0.98 (95)	0.47 (1371)
R	0.00 (8)	0.83 (41)	1.00 (21)	1.00 (141)	1.00 (166)	1.00 (26)	1.00 (327)
B	0.00 (40)	0.00 (168)	0.00 (33)	0.00 (132)	0.14 (43)	1.00 (10)	0.01 (282)

Notes: Entries are frequencies of voting for Red; observation counts are in parentheses. R and B denote red and blue, respectively; "–" denotes no signal in the row labels and no message in the column labels.

D.6 Additional results on group accuracy and welfare

Full-aggregation decomposition. Table 11 reports the decomposition of the communication effect on group accuracy described in Section 4.3 (Panel A), together with the analogous decomposition for average payoff (Panel B). For average payoff, the full-aggregation

benchmark replaces the observed group decision with the Bayesian-optimal decision based on the signals actually acquired, while holding acquisition choices and their costs fixed. At the low cost, 1.73 percentage points of the 8.05-point payoff gain are attributed to acquisition, net of information costs, and 6.32 points to aggregation. At the high cost, the corresponding contributions are 3.38 and 4.00 points.

Table 11: Decomposition of the communication effect on group outcomes

	Low cost	High cost
Panel A. Group accuracy		
Acquisition	2.71	5.28*
	(2.03)	(3.07)
Aggregation	6.32***	4.00
	(1.52)	(2.43)
Total effect	9.03***	9.28***
	(2.69)	(3.31)
Panel B. Average payoff		
Acquisition, net of information costs	1.73	3.38
	(1.58)	(2.10)
Aggregation	6.32***	4.00
	(1.52)	(2.43)
Total effect	8.05***	7.38**
	(2.34)	(2.65)

Notes: Entries are percentage-point effects of communication, estimated from group-round-level regressions on a communication indicator. The acquisition component is the change in the full-aggregation benchmark; the aggregation component is the reduction in the gap between the benchmark and the observed outcome; the two sum to the total effect. Because acquisition choices and costs are held fixed in the payoff benchmark, the aggregation component is identical for both outcomes. Standard errors, clustered by matching group, are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Alternative counterfactual decomposition. As a robustness check, I use an alternative decomposition based on the number of signals acquired by a group, k. For each treatment, let the acquisition component be represented by the empirical distribution of k, and the aggregation component by the mean outcome conditional on k. The forward decomposition first replaces the no-communication distribution of k with the communication distribution while holding conditional outcomes fixed, and then replaces the conditional outcomes. The reverse decomposition changes these objects in the opposite order, and the Shapley decomposition averages the contributions across the two orderings.

Table 12 reports the results. Both acquisition and aggregation contribute positively to the gains in accuracy and average payoff at both cost levels, although their precise attribution varies with the ordering.

Table 12: Alternative decompositions of group outcomes

	Low cost			High cost		
	Forward	Reverse	Shapley	Forward	Reverse	Shapley
Panel A. Group accuracy						
Acquisition	4.68*	3.34	4.01*	3.91*	6.07*	4.99*
	(2.51)	(2.30)	(2.32)	(2.18)	(3.44)	(2.77)
Aggregation	4.35**	5.69**	5.02***	5.37**	3.21	4.29*
	(1.76)	(2.27)	(1.93)	(2.22)	(2.77)	(2.38)
Panel B. Average payoff						
Acquisition	3.70*	2.36	3.03	2.01*	4.16*	3.09*
	(2.04)	(1.88)	(1.86)	(1.11)	(2.44)	(1.72)
Aggregation	4.35**	5.69**	5.02***	5.37**	3.21	4.29*
	(1.75)	(2.26)	(1.92)	(2.23)	(2.79)	(2.40)

Notes: Entries are percentage-point contributions. Standard errors are from 10,000 matching-group cluster bootstrap replications. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

D.7 Additional diagnostic evidence

This appendix reports subjects' beliefs about other members' acquisition and the corresponding realized frequencies at each cost level, together with the conditional choice data and the empirical value of acquisition by contingency in the high-cost treatments.

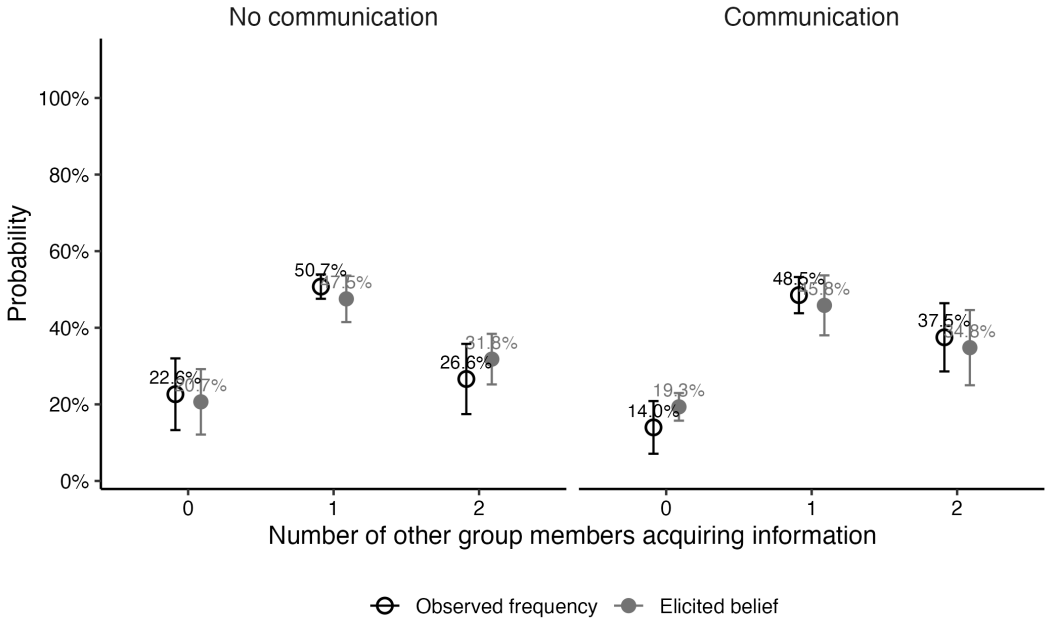


Figure 9: Beliefs about others' information acquisition and observed frequencies at the low information cost. Points show means across matching groups; error bars indicate 95% confidence intervals based on matching-group means.

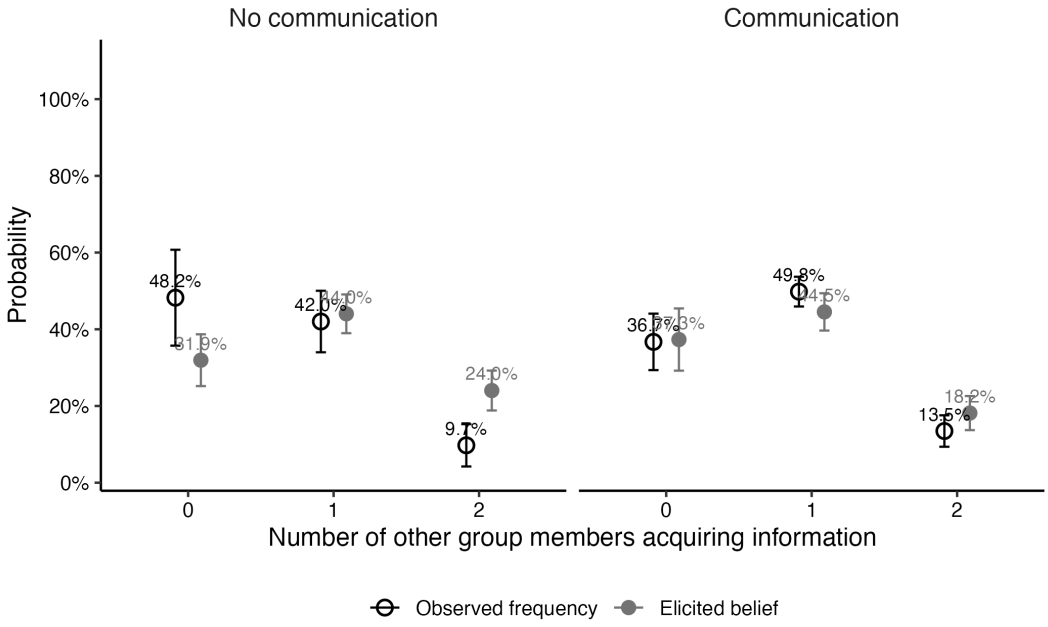


Figure 10: Beliefs about others' information acquisition and observed frequencies at the high information cost. Points show means across matching groups; error bars indicate 95% confidence intervals based on matching-group means.

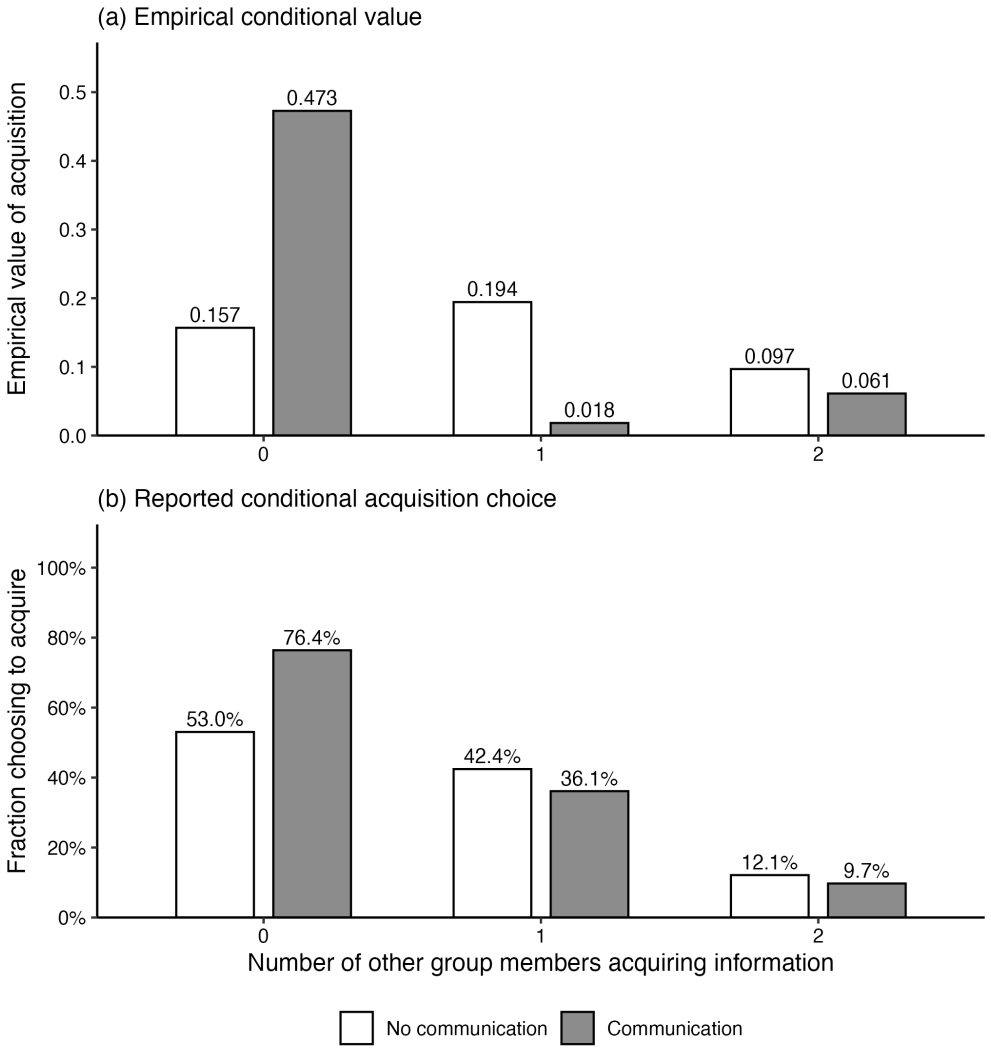


Figure 11: Conditional value of acquisition and reported acquisition choices at the high information cost.

E Alternative Explanations and Level Departures

This appendix supplements the discussion in Section 5. For brevity, I denote the four treatments by LNC, LC, HNC, and HC, where L/H indicates the information cost and NC/C whether communication is unavailable or available. I first consider whether the missing substitution effect can be accounted for by alternative equilibrium selection, departures from the benchmark preference assumptions, or bounded strategic reasoning. I then turn to the broader quantitative departures of observed acquisition rates from the equilibrium predictions and examine the extent to which they can be captured by quantal response. Overall, these alternatives have limited or partial explanatory power: some can account for particular features of the aggregate acquisition pattern, but none explains the full set of evidence, especially the conditional-choice results.

E.1 Alternative explanations

Equilibrium selection. One possibility is that subjects do not select the full-revelation and sincere-voting post-acquisition equilibrium on which I focus. As shown in Section 4.2, however, communication was highly informative and voting was largely sincere in the experiment. Moreover, moving toward a less informative communication equilibrium would attenuate both the persuasion and substitution effects, rather than naturally eliminating only the latter. A second possibility is that subjects coordinate on an asymmetric equilibrium at the acquisition stage. As shown in Appendix C.1.3, however, asymmetric pure-strategy equilibria imply the same two-sided treatment pattern as the symmetric equilibrium and therefore do not resolve the qualitative discrepancy. The data also provide little evidence of coordination on asymmetric acquisition roles: within each treatment, the distribution of the number of acquirers per group is statistically indistinguishable from a Binomial$(3, \hat{p})$ distribution, where $\hat{p}$ is the treatment-level acquisition rate (one-sample Kolmogorov–Smirnov tests with Holm adjustment; all adjusted $p = 0.57$).

Risk and social preferences. A second class of explanations relaxes the benchmark preference assumptions. Risk preferences cannot account for the missing substitution response. Consider the contingency in which exactly one other voter acquires information. With communication, that voter's signal is transmitted to the group, and acquiring a second signal does not improve the probability of a correct collective decision; it only adds the acquisition cost. Without communication, by contrast, the informed voter's signal can be offset by the uninformed voter's random vote, so an additional signal still improves the probability of

a correct majority decision. It follows that, for any increasing utility function over monetary payoffs, communication strictly lowers the expected-utility return to acquisition in this contingency. Risk attitudes may change the overall attractiveness of acquisition but not the direction of this communication effect. The conditional-choice data, however, show no corresponding decline in acquisition when communication is introduced. Standard inequality aversion leads to the same conclusion. Conditional on one other voter acquiring, the payoff-inequality consequences generated by acquisition costs are the same in the communication and no-communication games; communication changes only the instrumental value of the additional signal. Hence, under Fehr–Schmidt preferences, the utility return to acquisition remains lower under communication. Altruism likewise preserves this ordering, since it places additional weight on the common benefit of a correct decision without restoring any instrumental value to a second signal under communication. These preference departures can therefore shift overall acquisition incentives, but they do not naturally explain the absence of the predicted substitution response.

Level-k reasoning. A different possibility is that subjects engage in limited strategic reasoning about others' acquisition decisions. On the one hand, level-k reasoning can help explain why the predicted negative low-cost effect is absent while the positive high-cost effect can survive. Table 13 reports the predictions of a simple level-k model under the experimental parameterization $N = 3$ and $q = 0.9$. Starting from a nonstrategic anchor acquisition rate $\alpha_0 \in \{0, \frac{1}{2}, 1\}$, each subsequent level best responds to the acquisition probability generated by the preceding level. Across all three anchors and reasoning levels, communication does not change the acquisition choice at the low cost: $LC = LNC$. At the high cost, communication weakly raises acquisition, with $HC > HNC$ for some levels and equality for the others. Hence, any mixture over levels predicts no communication effect at the low cost and a weakly positive effect at the high cost. It can therefore eliminate the predicted negative low-cost effect and generate the positive high-cost effect, although it does not reproduce the positive low-cost treatment effect in the data.[39]

On the other hand, and more importantly, level-k reasoning does not account for the conditional-choice evidence. When subjects are explicitly told that exactly one other voter acquires, no strategic inference about others' acquisition behavior is required, yet the pre-

[39]The intuition is that acquisition best responses are bang-bang. At the low cost, for the beliefs visited by the iteration, the acquisition returns in the communication and no-communication games lie on the same side of the cost threshold, so communication does not change the acquisition choice even when it changes the underlying return. At the high cost, by contrast, the persuasion effect at zero acquisition crosses the cost threshold: $\Delta^C(0) = 0.4 > 0.25 > 0.2 = \Delta^{NC}(0)$. The treatment difference therefore survives for levels that best respond to zero acquisition.

Table 13: Level-k acquisition choices

Anchor	Level	LNC	LC	HNC	HC
$\alpha_0 = 0$	odd	1	1	0	1
	even	0	0	0	0
$\alpha_0 = \frac{1}{2}$	odd	1	1	0	0
	even	0	0	0	1
$\alpha_0 = 1$	odd	0	0	0	0
	even	1	1	0	1

Notes: Entries report the acquisition choice of a level-k voter under the experimental parameters, with 1 denoting acquisition and 0 no acquisition. "Odd" and "even" refer to odd and even $k \geq 1$.

dicted decline in acquisition under communication remains absent.

E.2 Quantitative departures and quantal response

Beyond the missing substitution effect, observed acquisition rates also depart from the symmetric equilibrium predictions in levels. These departures mostly take the form of more interior behavior: acquisition in LNC is substantially below its equilibrium prediction (0.520 versus 0.883, $p < 0.001$), while acquisition in both high-cost treatments is above the corresponding predictions (0.308 versus 0 in HNC, $p < 0.001$; 0.384 versus 0.215 in HC, $p = 0.003$). Acquisition in LC does not differ significantly from its equilibrium benchmark at the 5% level (0.618 versus 0.560, $p = 0.077$).[40]

I therefore estimate a symmetric logit-QRE model, together with extensions allowing for the preference and inference departures considered above. In each specification, the predicted acquisition rate in game X is the symmetric fixed point

$$\alpha^X = \frac{\exp\{\lambda D^X(\alpha^X)\}}{1 + \exp\{\lambda D^X(\alpha^X)\}},$$

where $D^X(\alpha)$ is the specification-specific utility return to acquisition net of its cost and $\lambda \geq 0$ measures response precision. I estimate the parameters by minimum distance on the four treatment-level acquisition rates and use the root mean squared error (RMSE) across treatments as a descriptive measure of fit. In addition to the baseline QRE, I consider CARA utility, one-parameter inequality aversion, and cursed message inference as introduced in Section 5.2.

For the cursed-message-inference specification, I use the conditional perceived acquisition values $\widetilde{\Delta}_k^C(\chi)$, $k \in \{0, 1, 2\}$, characterized in Section 5.2. Given that each other voter

[40]The p-values are from two-sided one-sample Wilcoxon signed-rank tests of matching-group average acquisition rates against the equilibrium prediction.

acquires independently with probability α, the perceived value of acquisition under communication is

$$\widetilde{\Delta}^C(\alpha,\chi) = (1-\alpha)^2\widetilde{\Delta}_0^C(\chi) + 2\alpha(1-\alpha)\widetilde{\Delta}_1^C(\chi) + \alpha^2\widetilde{\Delta}_2^C(\chi).$$

Cursed message inference affects only game C; acquisition incentives in game NC remain as in the benchmark.

Table 14: Quantal-response fits of information acquisition

Specification	Estimated parameters	LNC	LC	HNC	HC	RMSE
Observed	–	0.520	0.618	0.308	0.384	–
Nash equilibrium	–	0.883	0.560	0.000	0.215	0.254
Logit QRE	$\hat{\lambda} = 7.556$	0.600	0.521	0.374	0.359	0.072
QRE + CARA	$(\hat{\lambda}, \hat{A}) = (13.620, 5.130)$	0.551	0.533	0.295	0.435	0.052
QRE + inequality aversion	$(\hat{\lambda}, \hat{\eta}) = (4.426, 0.999)$	0.580	0.518	0.349	0.343	0.065
QRE + cursed message inference	$(\hat{\lambda}, \hat{\chi}) = (8.970, 0.282)$	0.614	0.599	0.356	0.406	0.055

Notes: Entries in columns LNC–HC are observed or predicted acquisition rates. Parameters are estimated by minimum distance on the four treatment-level acquisition rates. RMSE is the root mean squared deviation between predicted and observed treatment-level rates. The CARA specification uses $u_A(x) = (1 - e^{-Ax})/A$. The inequality-aversion specification imposes the one-parameter symmetric restriction $U_i = x_i - \frac{\eta}{2}\sum_{j\neq i}|x_i - x_j|$. In the cursed-message-inference specification, $\tilde{q}(\chi) = (1-\chi)q + \chi/2$ is the perceived precision of another informed voter's message.

The baseline QRE substantially reduces the quantitative discrepancy in levels, lowering the RMSE from 0.254 under the Nash benchmark to 0.072. As expected, noisy best responses pull the relatively extreme equilibrium predictions toward the interior. This improvement in levels, however, does not resolve the treatment-effect pattern. The fitted model predicts that communication lowers acquisition by 7.9 percentage points at the low cost and by 1.5 percentage points at the high cost, whereas the observed effects are increases of 9.8 and 7.6 percentage points, respectively. Thus, choice noise alone can account for part of the tendency toward interior acquisition, but not for the response to communication observed in the experiment.

Allowing for risk aversion improves the aggregate fit further. The CARA specification predicts an almost zero communication effect at the low cost and a positive effect at the high cost, with an RMSE of 0.052. Doing so, however, requires very strong curvature, $\hat{A} = 5.13$: the implied certainty equivalent of an equal-probability lottery paying either 1 or 0 is only about 0.13. More importantly, as discussed above, risk attitudes cannot account for the conditional-choice evidence. Conditional on exactly one other member acquiring, communication lowers the expected-utility return to acquisition for any increasing utility function, whereas the corresponding decline is absent from subjects' reported choices.

Inequality aversion performs less well. The estimated parameter is essentially at the upper boundary, $\hat{\eta} = 0.999$, yet the model continues to predict a negative communication

effect at both cost levels. Allowing for strong payoff-inequality concerns therefore does little to account for the qualitative discrepancy.

Cursed message inference produces a different pattern. The estimate $\hat{\chi} = 0.282$ implies partial under-inference from others' messages. The fitted low-cost communication effect is only -1.5 percentage points, so the large negative effect predicted by the benchmark is substantially attenuated, while the fitted high-cost effect is positive at 5.0 percentage points. Its overall fit, with an RMSE of 0.055, is close to that of the CARA specification. Thus, cursed message inference does not reproduce the positive low-cost communication effect in the data, but it captures the central asymmetry substantially better than the baseline QRE: the substitution effect is strongly attenuated while the positive high-cost effect is preserved. This pattern is also consistent with the conditional-choice evidence in Section 5.1 and the under-inference interpretation developed in Section 5.2.

Taken together, quantal response helps account for the tendency toward more interior acquisition rates, but choice noise alone does not explain the treatment pattern. Preference-based extensions improve some aspects of the aggregate fit but do not account for the conditional-choice evidence. Cursed message inference provides a more direct partial account of the missing substitution response, although it does not fully reproduce the positive low-cost communication effect observed in the data.

F Experimental Instructions

This appendix reproduces the experimental instructions in consolidated form, with treatment-specific text in square brackets. Subjects read the instructions on screen and also received a printed copy of the Part 1 instructions.

F.1 General Instructions

Welcome. Thank you for participating in this experiment. Please take a few moments to read the instructions carefully.

During the experiment, please do not talk to other participants. If you have any questions, raise your hand, and an experimenter will assist you.

The experiment consists of two parts. In Part 1, you will play a game for 30 rounds. In Part 2, you will complete three short tasks.

All earnings during the experiment will be measured in points, with a conversion rate of

100 points = 1 euro. You will receive a base payment of 650 points, which includes your participation fee. Your final payment will be

Your base payment + Part 1 earnings + Part 2 earnings.

In each part, your earnings will depend on decisions made by you and the other participants. The method for calculating payoffs will be explained in the instructions for each part. Your final payment will not be lower than 400 points.

All decisions made during the experiment are anonymous. Participants will not know one another's identities.

F.2 Instructions for Part 1

Please read the instructions carefully. After reading them, you will be asked to answer a few comprehension questions. You must answer all questions correctly in order to proceed.

Overview. Part 1 consists of 30 rounds, all following the same rules. At the end of the experiment, one round will be randomly selected for payment. You will not know in advance which round will be chosen.

In each round, you will be assigned to a group of three participants. Your group's task is to guess which of two jars has been selected. If your group guesses correctly, each member will receive 1000 points. A round proceeds as follows: first, your group is formed and the correct jar is determined by the computer. Then, you will have the opportunity to gain information about the correct jar. [In the communication treatments: After that, you can send a message to the group.] Next, you will vote, and your group's guess will be determined by majority rule. At the end of the round, the correct jar is revealed and your earnings are calculated.

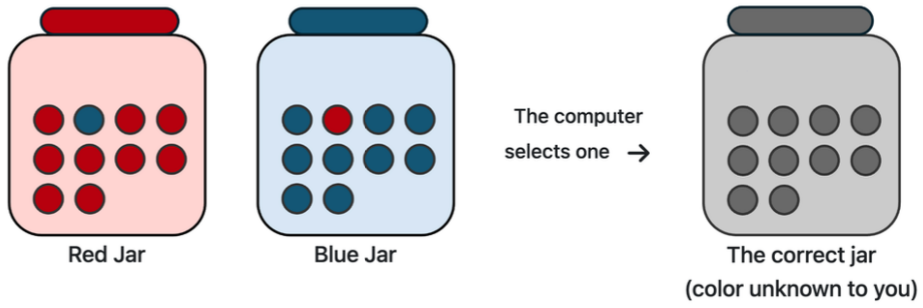
Group Formation. At the beginning of each round, the computer will randomly match you with two other participants to form a group. You will not know the identities of your group members. In each round, you will be rematched with different participants.

The Correct Jar. There are two jars: the Red Jar and the Blue Jar. The Red Jar contains 9 red balls and 1 blue ball. The Blue Jar contains 9 blue balls and 1 red ball.

At the beginning of each round, the computer randomly selects one of the two jars to be the correct jar for your group. Each jar has an equal chance of being selected: 50% for the Red

Jar and 50% for the Blue Jar. You and your group members will not be informed which jar is correct until the end of the round.

The correct jar is re-selected at the start of each round, independently from other rounds. This means that the chance of your group's correct jar being the Red Jar or the Blue Jar is not affected by what happened in previous rounds. The computer's choice is always completely random in each round, with both jars equally likely to be chosen.

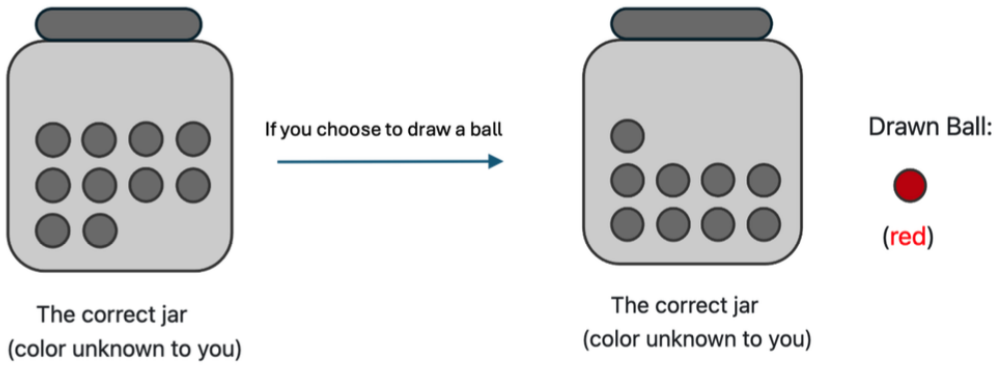


Drawing a Ball. After the correct jar is selected by the computer, each group member will have the opportunity to gain information by choosing whether or not to draw a ball from the correct jar. You will be asked to choose between the following two options:

- Draw a ball (cost: [100/250] points).
- Do not draw a ball (cost: 0 points).

If you choose to draw a ball, [100/250] points will be deducted from your earnings for the round. If you choose not to draw a ball, there is no cost.

If you choose to draw a ball, the computer will randomly select one ball from the correct jar on your behalf. The color of the ball will then be displayed on your screen. An example:



Balls are drawn independently for each group member who chooses to draw. This means that the color of your ball is not affected by whether other group members draw a ball or which ball they draw.

Your decision to draw a ball, or not, is private. You will not be informed whether other group members chose to draw a ball, and they will not be informed of your decision.

Sending a Message (communication treatments only). After all group members have made their drawing decisions, and those who drew have seen the color of their balls, each member will have the opportunity to send a message to the group, regardless of whether he or she chose to draw.

You will be asked to choose one of the following three options:

- Send "I have drawn a red ball."
- Send "I have drawn a blue ball."
- Do not send a message.

There is no cost associated with any of these options.

Once all group members have made their choices, each member will see all messages (if any) that were sent by group members. For example, if you choose to send "I h ave drawn a blue ball.", another member sends "I h ave drawn a red ball.", and the third member chooses not to send a message, then the following will be displayed on your screen:



Voting. [In the communication treatments: After the message-sending stage and the display of any messages sent,] each group member will be asked to choose between the following two options:

- Vote for the Red Jar.

- Vote for the Blue Jar.

Abstention is not allowed.

Your group's final decision will be made by majority rule. That is, if two or three members vote for the Red Jar, your group's decision will be the Red Jar. If two or three members vote for the Blue Jar, your group's decision will be the Blue Jar.

Once all group members have submitted their votes, the computer will count them and display the number of votes for each jar on your screen, along with your group's final decision.

Round Earnings. Your earnings in a round are calculated as

$$\text{Earnings} = \text{Benefits} - \text{Costs}.$$

Your benefits depend on whether your group's decision is correct—that is, whether it matches the correct jar. If the group's decision is correct, each member receives 1000 points. If the group's decision is incorrect, each member receives 0 points.

Your costs depend on whether you chose to draw a ball. If you did, [100/250] points will be subtracted from your earnings for that round. Specifically:

- If the group decision is correct and you did not draw a ball, your earnings are 1000 points.
- If the group decision is correct and you did draw a ball, your earnings are [900/750] points.
- If the group decision is incorrect and you did not draw a ball, your earnings are 0 points.
- If the group decision is incorrect and you did draw a ball, your earnings are [−100/−250] points.

The round earnings of the other two group members will be computed in the same way.

Results and History. After your group's decision has been made and you click "See the results," the correct jar will be revealed, and your earnings for the round will be calculated and displayed. Your earnings will remain private and will not be shown to other participants. Likewise, you will not be informed of other participants' earnings.

You will not be informed whether other group members chose to draw a ball, nor will they be informed of your decision to draw or not.

At any stage of the game, you can click the "Show History" button to review what happened in previous rounds and click "Hide History" to close the history display.

Next Steps. After reading the instructions, you will proceed to the comprehension questions. You must answer all questions correctly before proceeding. You will then play one practice round, which will not be selected for payment, followed by the 30 formal rounds.

In the final round and immediately afterward, you will be asked some additional questions that will not affect your payoff. After that, you will move on to Part 2 of the experiment.

F.3 Screens for Additional Questions

Screens 1–3 appeared in sequence before the decisions of round 30. After answering these questions, subjects played round 30 in the usual way. Screen 4 appeared immediately after round 30.

[Screen 1]

Before making your decisions in this round, you will be asked to answer a few questions.

Your answers to these questions will **not** affect your earnings in the experiment. However, we kindly ask that you respond as accurately and honestly as possible — your responses are valuable to our research.

Afterward, you will proceed with the game as usual. Earnings from this round can still be randomly selected for payment, **just like all other rounds**.

Click "Next" to begin answering the questions.

[Screen 2]

Q1: How likely do you think other group members will draw a ball?

Please use the sliders below to indicate how likely you think each outcome is. The total must add up to 100%.

Probability that 0 other members will draw a ball:

 36%

Probability that 1 other member will draw a ball:

 24%

Probability that 2 other members will draw a ball:

 8%

Remaining: 32%

[Screen 3]

Q2: If you choose to draw a ball, how likely do you think your group will make the correct decision in this round? (enter 0–100%)

Q3: If you choose not to draw a ball, how likely do you think your group will make the correct decision in this round? (enter 0–100%)

[Screen 4]

You have completed all the 30 rounds of Part 1. Please answer several questions below.

If you chose to draw a ball in some rounds, what were your top two reasons for doing so? *(Select up to two)*

"I wanted to know more about the correct jar myself before voting" / "I wanted to influence or lead others' votes" / "I wanted to be able to share something when sending a message" / "I followed what others did" / "I felt responsible for contributing to the group" / "I didn't really think about it" / "Other (Please specify)"

If you chose not to draw a ball in some rounds, what were your top two reasons for doing so? *(Select up to two)*

"I could learn about the jar from others' messages; no need to buy it myself" / "Others who drew would likely vote correctly anyway" / "I wasn't sure if drawing would actually help" / "I didn't want to risk earning negative points" / "I wanted to decide quickly without extra steps" / "I didn't really think about it" / "Other (Please specify)"

If you knew how many of other members chose to draw a ball, would you choose to draw a ball?

- If neither of the other two members draws, would you draw?
  "Yes, I would draw," "No, I would not," "Not sure."
- If one of the other two members draws:
  "Yes, I would draw," "No, I would not," "Not sure."

- If both of the other two members draw:
  "Yes, I would draw," "No, I would not," "Not sure."

F.4 Instructions for Part 2

Part 2: Three Tasks. Thank you for completing Part 1 of the experiment. We will now begin Part 2.

This part consists of three short, independent tasks. They will cost you no more than 10 minutes. At the end of the experiment, only one of these three tasks will be randomly selected for payment. Your earnings from the selected task will be added to your base payment and your payoff from the selected round of Part 1.

You will not know in advance which task will be selected before the end of the experiment, and you will not be told your earnings from the unselected tasks.

Each task will be explained before it begins. When you are ready, click "Next" to begin Part 2.

Task 1: Risk Attitude. You have 100 points. You can choose how many points to invest in a risky option, and how many to keep.

- Points you keep will be added directly to your payoff.
- Points you invest have a 50% chance to be lost, and a 50% chance to be multiplied by 2.5.

Please enter how many points you want to invest in the risky option (min. 0, max. 100).

Task 2: Strategic Uncertainty. In this task, you are randomly assigned to a group of 3 participants.

You will make a total of 10 decisions between Option A and Option B, displayed as 10 rows in the table below.[41] For each row:

- If you choose A, you will receive the indicated number of points for sure.

[41] The table listed 10 rows. Option A yielded a sure payment increasing from 10 to 100 points in increments of 10; Option B was the same in every row.

- If you choose B, your payment depends on how many group members choose B: if at least 2 out of 3 group members (including you) choose B, you receive 100 points; if fewer than 2 group members (including you) choose B, you receive 0 points.

One of the 10 rows will be randomly selected to determine your earnings for this task. Your earnings will depend on your choice and the choices of the other group members in that row. Note: If you select Option A in any row, all subsequent rows will automatically be set to Option A. You can still revise your choices by adjusting where you first select Option A.

Task 3: Social Value Orientation. In this task, you will be randomly paired with another participant.

On the following pages, you will make decisions in six different situations, each involving how to allocate points between yourself and the other participant.

After both of you have made all six decisions, one of the six situations will be randomly selected to determine your payment for this task. You and the other participant will be randomly assigned the roles of Decision Maker and Receiver.

- If you are assigned the role of Decision Maker, your choice in the selected situation will determine both your own and the other participant's payoff for this task.
- If you are assigned the role of Receiver, your payoff will depend on the Decision Maker's choice.

Please click "Next" below to start the task.

Example allocation screen (Task 3).[42]

Table 15: Allocation options in the Social Value Orientation task

Situation	Own payoff	Other's payoff
1	100, 94, 88, 81, 75, 69, 63, 56, 50	50, 56, 63, 69, 75, 81, 88, 94, 100
2	85, 87, 89, 91, 93, 94, 96, 98, 100	15, 19, 24, 28, 33, 37, 41, 46, 50
3	50, 54, 59, 63, 68, 72, 76, 81, 85	100, 98, 96, 94, 93, 91, 89, 87, 85
4	50, 54, 59, 63, 68, 72, 76, 81, 85	100, 89, 79, 68, 58, 47, 36, 26, 15
5	85, 85, 85, 85, 85, 85, 85, 85, 85	85, 76, 68, 59, 50, 41, 33, 24, 15
6	100, 98, 96, 94, 93, 91, 89, 87, 85	50, 54, 59, 63, 68, 72, 76, 81, 85

You receive:	100	94	88	81	75	69	63	56	50
	○	⊙	○	○	○	○	○	○	○
Other receives:	50	56	63	69	75	81	88	94	100

[42]The six situations are reproduced in Table 15.

If this situation is selected and you are the Decision Maker:

You will receive **94** points.

The other participant will receive **56** points.

F.5 Post-Experiment Questionnaire

After completing Part 2, subjects reported their age, gender, nationality, and current level of education and were given an opportunity to provide comments or feedback about the experiment. The computer then revealed the randomly selected round from Part 1 and the randomly selected task from Part 2 and displayed the subject's total experimental earnings.